LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

A Capstone Project presented to the
College of Computer Studies of
Pamantasan ng Lungsod ng Pasig

In partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

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ABSTRACT

Title : LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy

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Project Description:

LearnSync AI is a web-based learning companion that enables Junior High School students of Pasig Christian Academy to learn independently with only their teachers' materials. It uses Bayesian Knowledge Tracing (BKT) to estimate understanding of each concept, plus Generative Artificial Intelligence to construct tests, games, and feedback. With a quantitative developmental research design, the study found that 62 surveyed students strongly agreed on the need for a companion that tracks their progress (composite mean = 3.38). The BKT engine computed all 12,067 updates correctly while 36 out of 37 Generative AI outputs passed verification. After one week of use, the mean knowledge estimate of 46 students increased from 34.5% to 75.3% ($p < .001$), which indicates recorded progress but does not confirm learning gain. Fifty-seven evaluators found the companion acceptable under ISO/IEC 25010:2011 (mean = 3.08, Agree), with only Performance Efficiency rated Disagree (1.85). Improving response speed and pre test and post test is recommended.

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

The rise of digital learning has enabled independent learning to become an essential pillar of modern pedagogy. Students engage in independent learning activities such as review, flashcards, and quizzes to complement their learning beyond the classroom, with some applications providing a one-size-fits-all approach to reviewing topics. Contemporary studies argue that personalized learning is a paradigm shift that moves away from generic instruction toward contextualized and learner-centric learning (Fariani et al., 2023; Wang et al., 2025). As such, students may find themselves studying concepts they already understand while struggling with topics that require additional support, and activities that do not account for a learner's level of difficulty may lead to disengagement and boredom (Krannich et al., 2025).

Cognitive Load Theory posits that the key to successful learning is optimizing the presentation of information to prevent the working memory from being overwhelmed with unnecessary information. In digital learning, text-heavy digital interfaces and multimedia can cause extraneous cognitive load, which ultimately offsets the learning process and decreases cognitive performance (Krieglstein et al., 2023; Surbakti et al., 2024). Personalized learning considers these cognitive factors and is a more effective instructional strategy than a uniform presentation of learning materials since students differ in terms of prior

knowledge, attention, and literacy in navigating digital interfaces (Skulmowski & Xu, 2022; Surbakti et al., 2024).

Digital learning technologies have provided students with enhanced accessibility to independent learning activities outside formal instruction by allowing them to learn at their own pace and set their own learning schedule. Additionally, independent learning has been enabled through learning assessments and other tools that promote engagement and participation (Alshammary & Alhalafawy, 2023; Josué et al., 2023; Sisouvong & Pasanchay, 2024). However, most applications only offer a singular independent learning activity at a time, thereby requiring students to use multiple applications to engage in their review activities (Qiu & Qiu, 2026).

The use of multiple independent learning activities simultaneously was found to increase an application's unrelated cognitive load and tool fatigue, which suggests the need to consider an integrated independent learning platform. An integrated learning system can unite independent learning activities and analytics to track progress and engagement, and studies report that an integrated learning environment has been found to increase learning effectiveness and motivation compared to independent learning (Du et al., 2025; Elmabaredy & Gencel, 2024; Yang et al., 2024). Finally, learning analytics can be used to improve independent learning by enabling students to monitor their learning performance, detect learning gaps, and update their learning strategies, especially when combined with instructional support (Kleimola et al., 2024).

AI-driven independent learning further enhances student engagement by providing each learner with a personalized learning experience, learning resources, and feedback based on their learning performance. Unlike conventional applications, AI can process the learning data of each student to determine what topics they may need additional support with and recommend learning resources and activities to address such learning gaps (Barrera Castro et al., 2024). As such, research argues that AI-driven independent learning can be an effective tool in education by improving the learning experience through adaptive support based on each learner's performance, enabling students to operate at their own pace while still providing much-needed guidance (Bayly-Castaneda et al., 2024).

Pasig Christian Academy is a private school established in 1986 that offers Preschool, Elementary, and Junior High School education in Pasig City. This paper will focus on the Junior High School department as students in this department begin to learn how to practice independent learning as preparation for their Senior High School education. Based on the researcher's requirements analysis and discussions with the school administration, independent learning at PCA is mainly reliant on official textbooks and teacher-provided materials. Such learning materials are sufficient in supporting the school's curriculum; however, they present a uniform presentation regardless of a student's performance and level of understanding. The researchers' needs assessment uncovered that students often struggled to review topics they had difficulties understanding and wanted more options for independent learning activities. As such, students

expressed a need for independent learning tools that could track their progress, provide them with engaging activities, and offer an all-in-one independent learning platform.

Rather than eliminating such independent learning activities, the identified needs can be addressed by implementing LearnSync AI as an independent learning platform that offers personalized learning resources, adaptive independent learning activities, AI-driven study strategies and review, and learning analytics. LearnSync AI is designed to provide students with an engaging and adaptive independent learning experience that works in tandem with PCA's existing official textbooks and teacher-provided materials. LearnSync AI integrates Bayesian Knowledge Tracing (BKT) to estimate a student's learning progress and provide adaptive independent learning activities while utilizing Generative Artificial Intelligence (Generative AI) to generate learning support based on the educator-uploaded materials. To ensure the integrity of PCA's curriculum and the quality of learning materials, only educators can upload resources and review activities that are appropriate for each grade level.

Within the integrated platform, students can access Adaptive Flashcards, Adaptive Test Mode, AI study strategies such as Feynman and Free Recall, AI learning assistant, and AI educational games. BKT continuously tracks a student's knowledge of a subject based on their performance in the previously attempted independent learning activities, and such performance data can be utilized in Adaptive Flashcards, Adaptive Test Mode, and AI Educational Games. Students can also monitor their learning progress within the platform, including

their learning streaks, knowledge level, and achievements in the form of badges. By integrating educator-approved learning resources and BKT and Generative AI, LearnSync AI can offer an engaging and adaptive independent learning experience for students while providing them with learning analytics so that they can update their independent learning activities and address their learning gaps.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive synthesis of local and international literature, research studies, and technological frameworks relevant to the development of LearnSync AI.

Modern Study Challenges and Diverse Student Learners

Developing an efficient digital learning platform requires acknowledging the fact that students differ from each other in terms of learning styles, processing information, and responses to them. Students differ from each other in terms of prior knowledge, academic ability, motivation, learning pace, and study habits; a single approach will not be efficient for all students (Barrera Castro et al., 2024; Gligorea et al., 2023). Studies on personalized education indicate that learning environments can become more efficient if they take into account students' characteristics and needs (Barrera Castro et al., 2024). Similarly, adaptive learning has been suggested as an alternative to traditional

one-size-fits-all learning approaches since it offers learning experiences that are suited to individuals' needs and progress (du Plooy et al., 2024; Gligorea et al., 2023).

In addition, researches on adaptive learning underline the fact that educational systems that are able to answer the needs of the learner and their progress can offer more engaging and effective learning (du Plooy et al., 2024; Gligorea et al., 2023). In the context of the growth of digital learning, educational systems are more and more supposed to provide a flexible and personalized learning process that answers the multiple needs of the students (Barrera Castro et al., 2024).

Fast and Slow Learners

Students vary in terms of their learning processes, comprehension abilities, and completion levels. Therefore, the approach to teaching and acquiring knowledge at school differs significantly among individuals. In a mixed-ability classroom, one student might easily grasp the information provided by the teacher during a specific topic; hence, they may find the learning process boring. However, other students may need more time, practice, and support to fully comprehend a specific topic. Since students learn at different paces, applying the same methodology to teach everyone is ineffective, which minimizes the possibility of providing equal educational opportunities to individuals, resulting in demotivated learners. Studies on differentiated instruction emphasize the

importance of flexible learning and various techniques applied in classrooms to ensure that students learn according to their abilities and speeds (Ziernwald et al., 2022).

Likewise, studies on educational differentiation show that differentiating instruction according to individual learner needs may increase participation, motivation, and achievement in mixed-ability classrooms (Eikeland & Ohna, 2022). These results emphasize the importance of differentiating learning experiences so that students can learn based on their own level instead of being forced to do the same activities at the same pace.

Digital Overload

The growing reliance on the use of digital technologies in education has created pedagogical environments where students depend on multiple digital tools and platforms to interact with learning materials, carry out assessments, submit assignments, and interact with instructors and peers. While they can ease access and increase flexibility, navigating different interfaces and constantly switching between different systems can add to the cognitive load of completing learning tasks, thus reducing cognitive capacity for comprehending, processing, and retaining new information, especially when having to split attention between learning material and the interface navigation (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research on digital learning further suggests that unnecessary complexity in digital environments can lead to extraneous cognitive load, referring to mental activity triggered by the design of the learning system rather than the learning material, which can reduce learning efficiency and make it more difficult for students to focus on instructional content (Nordin et al., 2026).

Further, studies on learning analytics stress the need to design organized and accessible digital learning environments that facilitate learners through their access to educational resources and activities with reduced barriers to learning (Qiu & Qiu, 2026). Together, these pieces of research bring out the importance of building integrated learning platforms that consolidate academic resources and learning activities to allow for a more organized, efficient, and learner-centered learning experience.

Junior High School Students

It has been observed that in the context of the Philippines, distance and modular learning have necessitated greater independence on the part of students, who must rely less on teacher guidance and more on managing their own learning experiences and time schedules.

Studies have shown that it is critical, in such a context, for students to practice self-regulation, since differences therein could result in poorer learning outcomes (Briones et al., 2023; Calamlam, 2023; Funa et al., 2023). Meanwhile, there are still a number of challenges which Filipino learners have had to

overcome in order to effectively engage in self-led learning, such as less access to reliable sources, poor understanding of modules, and a lack of coherence in the learning environment (Abejuela et al., 2023; Agayon et al., 2022; Bustillo & Aguilos, 2022).

These conditions point to the need for learning platforms that offer organized access to resources, adaptive support, and timely feedback. The study of educational technology also suggests that students' learning experiences can be improved if digital systems are designed to be accessible, interactive, well-facilitated, and responsive to learners' needs by using adaptive learning systems and other technology-based interventions to enhance engagement and performance (Batucan et al., 2022; Caliston 2025; Llacuna et al., 2026).

Thus, developing a learning platform for junior high school students is justifiable within the literature since this category of learners is still building their learning behaviors and can benefit from having an integrated learning tool that enhances accessibility, uses appropriate technology, and provides personalized support to meet learners' needs (Barrera Castro et al., 2024; Salva et al., 2022; Villarama et al., 2022).

Students differ in their abilities, pace, and approaches towards studying, which means that a universal set of materials is not entirely beneficial for learners. Overall, the reviewed articles discuss the effectiveness of adaptive learning and prove its importance by demonstrating how well it is aligned with students' individual learning paths. The discussed studies also indicate that

students often have to manage multiple sources separately, which puts additional cognitive load on learners in terms of organizing and maintaining the materials. Therefore, considering the complexity of independent learning at the Junior High School level, the population would benefit from adaptive instruction and immediate feedback, as well as the individual path recommendation and instant organization of materials, which LearnSync AI offers through the combination of Generative Artificial Intelligence and Bayesian Knowledge Tracing algorithms.

Analysis of Independent Study Tools and Research Gaps

Independent learning resources have become an essential component of contemporary education as they provide students with flexible opportunities to revisit educational material and learn outside the classroom. At the same time, the majority of such tools are designed to cater to specific learning objectives, which prompts learners to resort to multiple applications to address their diverse needs, resulting in a fragmented digital learning experience. Studies demonstrate that such multiplicity decreases students' cognitive load and tool fatigue, thus distracting them from the actual educational tasks and focusing on the means of their accomplishment (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research also supports that effective instructional design and interactive learning environments can lower cognitive load, which suggests that more integrated digital platforms would better support independent learning than singular, isolated tools (Nordin et al., 2026). These findings identify an important

research gap in current independent study tools, specifically those which are based on a single evidence-based learning function and lack adaptive support in a unified cognitive environment.

Fragmentation of Current Educational Applications

Educational applications have been widely adopted to facilitate independent learning by providing a broad range of tools such as flashcards, quizzes, note-taking features, and progression tracking. Unfortunately, most such applications are limited to a few specific fields of study, thus forcing students to use a variety of such tools for different areas. This leads to a fragmented digital learning experience in which the focus is divided between the actual educational content and the various technological tools. Recent studies have shown that the multi-platform approach to learning creates an increased extraneous cognitive load, tool fatigue, as well as a general inability to focus on the task at hand instead of the learning technology itself (Ibrahim et al., 2025; Qiu & Qiu, 2026).

Research also indicates that fragmented learning resources are often poorly organized, further decreasing learning efficiency and constraining knowledge building compared to a unified system (Hu & Liu, 2025). Supporting this, integrated resource designs have been associated with greater perceived coherence while also reducing cognitive load, suggesting that a more unified educational system is potentially better at supporting independent learning than non-unified single-purpose applications (Lou, 2025).

Limitations of Static Review Systems

Many educational applications still utilise static review systems providing the same materials to all learners with the same review activities despite differences in learners' pace, prior knowledge and current understanding (Bakri, 2025). This approach could prove to be time-consuming since traditional reviews often distribute the same materials in a uniform manner rather than targeting each learner's weaknesses (Okubo et al., 2023) Research on personalized learning, however, defines adaptive systems as those that adjust to changing learner needs and recommend resources based on learner characteristics, knowledge level and preferences (Abu-Rasheed et al., 2023; du Plooy et al., 2024).

Evidence also suggests that adaptive review may facilitate learning: an individualized review dashboard that recommended pages that were insufficiently understood caused a significantly higher increase in quiz scores than a broader review, while a larger review found that adaptive systems can often improve academic performance and engagement compared with non-adaptive environments (Bakri, 2025; du Plooy et al., 2024; Okubo et al., 2023).

Most of the study applications currently available focus solely on one aspect of education or provide an equal opportunity of learning any subject regardless of the existing pace and comprehension levels of the students. Both options create a problem of choice and require the individuals to pay more

attention to the program rather than the contents. Therefore, the literature provides an insight into the need for a combined strategy that uses activities, material organization and provision of necessary support as a single source. Learn Sync AI is suggested as the option that incorporates the generative artificial intelligence, Bayesian Knowledge Tracing, adaptive activities, AI-based tutoring, gamification elements and learning analytics into one product.

Learning Analytics and Student Motivation

Learning analytics has proven to be a critical educational technology because learner-facing dashboards offer students direct access to information about their progress, enabling them to track their performances and recognize what they still need to learn (Paulsen & Lindsay, 2024; Yang et al., 2024). At the same time, dashboards can help students take control of their education by providing them with progress visuals, supporting reflection, feedback, and next steps for their further study. In fact, experimental and quasi-experimental evidence suggest that student-facing dashboards can enhance learning by improving self-awareness, academic self-efficacy, engagement, and overall achievement when the technology is deliberately designed to motivate behaviors and regulate skills, not only provide a learner with information (Fleur et al., 2023; Yang et al., 2024).

It is also worth mentioning that studies support the idea that dashboards are most effective when they are based on motivational theory and encourage

students to make purposeful decisions, rather than a one-size-fits-all approach that fails to account for their progress (Lee & Kim, 2024).

Learning Progress

Visualizing learning progress promotes students' self-monitoring through its ability to make both their current standings and learning trajectory visible during the process, not only at the end of the graded activity (Paulsen & Lindsay, 2024; Ponomariovienė et al., 2025). The former is linked to the concept of progression in a competency-based framework, which is associated with feedback, reflection, and self-regulation (Ponomariovienė et al., 2025). Indeed, the regular receipt of feedback enables learners to gain an understanding of their achievements and the strategies to pursue to progress towards the expected learning outcomes.

Learner-facing dashboards have come to focus far less on teacher-oriented analytics than their impact on students' learning (Paulsen & Lindsay, 2024). A review identified monitoring, awareness, and comparison as the most frequent dashboard affordances, but more recent dashboards do include elements of feedback, reflection, and goal-setting to better support learning action (Paulsen & Lindsay, 2024).

Evidence from higher education further supports the notion that dashboards may enhance self-assessment system engagement. When tested on a cohort of 194 students, individuals provided with dashboard access showed increased interaction with a computerized classification testing system, as well as

higher levels of confidence, in comparison to peers who engaged with the same technology in the absence of a dashboard (Ifenthaler & Şahin, 2023). The dashboard in this trial included assessment-related metrics such as the number of mastered subject areas, correct and incorrect responses, and overall attempts, providing insight into how progress visualization can better equip students to engage with formative assessments throughout the course of a semester (Ifenthaler & Şahin, 2023).

Study Consistency and Behavioral Engagement

Study streaks and digital badges are examples of features that motivate students to learn continuously, develop good learning behaviors, and maintain their interest in the learning process. The former is a feature that keeps track of a student's continuous learning activity, and the latter is a type of badge that rewards the learner for achieving specific objectives. These gamified features boost motivation, engagement, learning progress, and continuous participation (Fauzi et al., 2025; Jun & Lucas, 2024; Sappaile et al., 2025).

It must also be noted that the benefits of self-monitoring and gamification of feedback in education have also been supported by recent evidence. Compared to the control group, participants who shared achievements through badges had increased self-regulated learning and a higher completion rate of 70% compared to the control group's 56%. Additionally, a broader literature review also supports the efficacy of incorporating badges, points, and feedback

in gamified educational experiences in promoting active learning, progress tracking, and continued engagement if implemented correctly (Jun & Lucas, 2024).

Overall, current evidence supports the notion that streaks and badges can facilitate sustaining participation, increasing engagement, and developing consistent independent learning behaviors, albeit with a need for appropriate design and potential mixed effects for certain participants (Adanyin, 2024; Balci et al., 2022; Thakarkar et al., 2025).

Time-on-Task and Self-Regulated Learning

Time-on-task information in digital learning environments is useful because it makes self-regulated learning behaviors visible and evaluable during learning processes and not after. According to clickstream evidence, online learners vary in terms of the time spent and patterns of use within different performance categories, which indicates that time is a significant predictor of student effort in learning activities (Cao et al., 2022). Generally, studies on self-regulated learning technologies report that monitoring, goal-setting, and self-assessment processes are the most common functions provided in online learning contexts to facilitate learners' regulation practices, which aligns with the purpose of time-on-task displays (Perez-Alvarez et al., 2022).

Recent dashboard studies, in turn, suggest that students may value analytics displays for observing study progress and appraising their learning

behavior, although their impact depends on clear reference points and inclusion into everyday learning processes (de Vreugd et al., 2024; Kleimola et al., 2024). Overall, the display of the time-on-task appears to foster learner awareness, responsibility, and more intentional study habits, but the intervention literature shows mixed effects in cases when analytics are not related to planning and reflection activities (Heikkinen et al., 2022).

Across the articles, the ability for students to view their own learning data motivates and facilitates managed practice by providing visibility into learning progress and patterns. The display of progress, time tracking features, and rewards such as streaks and badges provide learners with the ability to maintain consistent learning habits and reflect on their weaknesses independently, which is essential when learning without the support of instructors. LearnSync AI caters to this need by providing a dashboard for each student to track their knowledge progress, learning streaks, and rewards in the form of badges. By doing so, the application enables learners to hold themselves accountable based on their own progress and patterns in their learning.

Using Artificial Intelligence for Personalized Student Support

Technology can be used to improve personalized student service by providing instructions in a manner that fits individuals' needs rather than giving the same content to everyone. Using artificial intelligence to create an educational program focuses on students' characteristics to offer them relevant

information, fill their gaps in the learned material, and suggest appropriate activities based on the collected data (Barrera Castro et al., 2024; Murtaza et al., 2022).

Recent studies also suggest that adaptive feedback and intelligent tutoring systems can enhance engagement, conceptual knowledge, and satisfaction compared with static or traditional approaches, although effectiveness depends on design quality and subject context (Naseer & Khawaja, 2025; Sajja et al., 2023; Villegas-Ch. et al., 2025). At the same time, the literature consistently notes concerns about privacy, algorithmic bias, infrastructure, and the need to integrate AI with human teaching rather than replace it (Bahroun et al., 2023; Lin et al., 2023; Yarlagadda, 2025).

AI in Modern Classrooms

AI integration in modern classrooms has enabled the shift from a standard form of instruction to an adaptive learning environment that accommodates student-specific needs. In recent articles, implementations of AI in modern education are considered as a paradigm shift away from standardized and uniform approaches to learning, with AI-driven systems being able to analyze student data, incl. performance, preferences, and interactions, in order to offer adaptive and personalized content, instruction pace, and tools (Lin et al., 2023; Rane, 2024; Yarlagadda, 2025). Adaptive platforms and tutoring systems are powered by machine learning algorithms, natural language processing, and

learning analytics to offer feedback, recommendations, and personalized learning paths that contribute to a more personalized educational experience (Akintola et al., 2025; Gligorea et al., 2023; Murtaza et al., 2022).

Empirical studies also report gains in engagement and academic outcomes, where a well-designed AI system can lead to an improvement of 25% in grades, test scores, and engagement in a higher-education experiment and higher post-assessment scores in a mixture of educational settings (Naseer et al., 2024; Sari et al., 2024; Villegas-Ch. et al., 2025). Nonetheless, the literature is explicit with respect to the fact that successful classroom integration requires educator training, infrastructure, ethical safeguards, and keeping the use of AI as a compliment rather than a replacement for human teaching (Barrera Castro et al., 2024; Halkiopoulos & Gkintoni, 2024; Walter, 2024).

AI in Finding Student Learning Gaps

Artificial Intelligence has become an essential tool in recognizing students' learning gaps and offering individualized academic assistance. While conventional assessments frequently only identify areas of weakness during periodic tests, AI-driven adaptive learning systems may constantly evaluate students' performance and learning styles to discover weaknesses and offer personalized help right away (Akintola et al., 2025; Khanipoor & Karimian, 2024; Onesi-Ozigagun et al., 2024).

These systems can then suggest targeted learning activities, adjust difficulty levels, and provide personalized feedback to help students focus on areas they lack understanding in and enhance their comprehension while studying independently (Akavova et al., 2023; Gupta, 2024).

Artificial Intelligence can aid education by providing lessons that are tailored to the learner, adapting and focusing on where the student may be struggling, whereas a single lesson for the entire class may not do that. LearnSync AI uses this by modifying their activities as the student progresses, and to bring attention to the concepts that they might need help on, so that their independent learning is always on track.

Combining Gamification and Evidence-Based Learning Strategies

Gamification and evidence-based learning strategies seem to contribute to various aspects of learning in digital platforms, but in different directions. Gamification mostly facilitates motivation and engagement to the learning process while evidence-based learning strategies like retrieval practice, repeated practice, and feedback are more closely linked to desirable outcomes such as retention and conceptual learning (Andretta et al., 2025; Nylén-Eriksen et al., 2025; Trayco et al., 2025). A number of recent reviews and empirical studies highlight the fact that the most successful gamified learning interventions are those that integrate good game design principles with effective feedback strategies and opportunities for practice and retrieval rather than focusing on

peripheral aspects like rewards (Dahalan et al., 2023; Lee et al., 2025; Zatserkovnyi, 2025).

Emerging work further suggests that AI-enhanced gamified systems may enhance this integration further by adapting difficulty, feedback, and reward timing to learners' progress while supporting long-term retention and motivation (Aswini et al., 2026; Kassenkhan et al., 2025; Kumar, 2026).

Gamification Procedures

Gamification is an innovative teaching method that increases students' engagement and performance by incorporating games' stimulating features into the learning process. Recent research makes it clear that gamification has a positive effect on students' performance, engagement, and motivation in the educational setting (Dahalan et al., 2023; Dehghanzadeh et al., 2023; Hellín et al., 2023; Khaldi et al., 2023; Qudsi, 2024). When implemented in online learning, gamification allows students to actively review the material they have already learned while keeping their engagement high.

Research also suggests that gamification is most effective when the game elements are aligned with sound instructional design and feedback and not only used as a reward system or for entertainment purposes (Dehghanzadeh et al., 2023; Khaldi et al., 2023; Kode, 2025; Trayco et al., 2025). However, the literature also reports contradictions in its impacts with regard to learning

outcomes due to learner heterogeneity, novelty effects, and issues with the quality of implementation (Ratinho & Martins, 2023; Ukgoda, 2025).

Shared Mechanism

Crossword puzzles, matching cards, and timed trivia questions are similar in many ways, as they all share the same learning process where students are required to recall and retrieve information instead of simply re-reading it, and gamification studies suggest that such activities can effectively improve learning outcomes, engagement, and persistence (Dahalan et al., 2023; Monzon & Hays, 2025). Moreover, they all share key game characteristics, such as challenge, immediate feedback, and clear goals, which can support student engagement and involvement throughout the review process (Al-Khresheh, 2025). Crossword puzzles, in particular, have been shown to improve terminology recall, vocabulary retention, and student participation due to the necessity of retrieving definitions and spelling based on given hints (Hsu et al., 2023; Jamal et al., 2026; Ramadhani & Shofiyuddin, 2025). Similarly, matching card activities have been shown to follow the same pattern, as such activities require students to identify connections between given concepts, increasing retention, interest, and confidence levels (Yu et al., 2025).

Timed trivia is related to the other two since it also relies on fast recognition, repetitive practice, and instant feedback on the answers to enhance memory and focus that gamified trivia has been associated with (Al-Khresheh,

2025). In essence, the three games are different but achieve similar goals since crosswords enhance retention of terms and words, matching cards promote associations between related concepts, and timed trivia improves recognition and feedback-driven review (Hsu et al., 2023; Jamal et al., 2026; Yu et al., 2025).

Crossword

Crossword puzzles can be used as an effective teaching tool to help students learn and retain information. In contrast to standard teaching and reviewing practices, they allow students to engage in the active retrieval of the learned material and aid comprehension through association of terms with their definitions or specific clues.

Research indicates that crossword puzzles can improve students' understanding of concepts and promote better performance on objective assessments by promoting active recall during learning activities (Arnold et al., 2024). Indeed, the integration of crossword puzzles into online learning software can enhance concept retention and increase student engagement during independent work.

Matching Card Games

Matching card games are often used as a helpful interactive tool to engage students as they learn material. As they are playing the game, students

are forced to interact with the content on a deeper level because they have to match terms or ideas. This can help them learn the ideas better since they are actively recalling the words or ideas and associating them together instead of passively reading them.

A recent study found that using matching card-type learning material can significantly enhance a student's lexicon and learning outcome by making the learning process engaging and interactive (Febrianti & Shofiyuddin, 2025). Due to the fact that matching activities involve learners into repeated recalling and associating of related items, the inherent principles can be utilized to build foundational knowledge within the context of an electronic learning environment.

Timed Trivia Questions

Timed trivia questions represent a gamified learning activity, which inherently incorporates retrieval practice. The review questions are presented in a timed environment, which encourages active learning and engagement. Recent studies have demonstrated that gamified quiz tools can enhance student interest, engagement, participation, and learning outcomes in educational settings (Dahalan et al., 2023; Nadeem et al., 2023; Özdemir, 2024; Raje & Tamilselvi, 2024).

In addition, elements such as time limits for gameplay, scorekeeping, leaderboards, and instant feedback during a game appear to increase the level of involvement and learning if they are consistent with instructional objectives (Jaramillo-Mediavilla et al., 2024; My et al., 2024; Ningsih, 2025; Ratinho & Martins, 2023). In combination, these results suggest that timed trivia questions can be an effective method for encouraging engagement while reinforcing concepts that have been learned through repeated practice and feedback (Pellas, 2024; Zainuddin, 2023).

Retrieval Practices

Cognitive retrieval practices are an integral aspect of contemporary educational psychology, as they substantially increase the efficiency with which students are able to recall information that has been taught to them. Compared to conventional approaches to learning, such as reiterating and reading through material, retrieval-based learning encourages the formation of stronger memories and a deeper comprehension of concepts (Madan, 2023; Roelle et al., 2022; Serra et al., 2025; Thompson & Hughes, 2023).

Research across educational settings demonstrates that retrieval practice tends to yield better results than passive review tactics in terms of learning and delayed retention, and could also enhance transfer, problem solving, and metacognitive monitoring (Corral & Carpenter, 2025; Muzsnay et al., 2025; Trumble et al., 2023). Retrieval practice could be carried out by employing such

strategies as free recall, low-stakes testing, and structured explanation exercises such as the Feynman Technique, which promote active reconstruction of knowledge from memory (Linh & Oanh, 2025; Rajesh & Khan, 2024; Wellmann & Torres Skillicorn, 2023).

Feynman Technique

The Feynman Technique is a learning strategy that promotes developing a deep understanding of the concepts by explaining the material in one's own words. Explanation-based learning tasks encourage learners to produce coherent verbal statements that serve to clarify, interpret, and justify information that helps to unify the newly acquired knowledge with the prior understanding and generalize it (Fiorella, 2023). Instead of rote learning, such tasks demand reconstructing knowledge and, thus, can enhance comprehension and retention (Roelle et al., 2023; Roelle et al., 2022).

The Feynman Technique similarly promotes identifying gaps and re-visiting materials until learners can accurately explain a concept (Linh & Oanh, 2025; Rajesh & Khan, 2024). Notably, this approach ties into established research that generative activities primarily promote building coherent representations whereas retrieval practice predominantly facilitates their consolidation in memory (Obergassel et al., 2025; Roelle et al., 2022; Roelle et al., 2023). Recent experimental evidence also shows that combining both generative and retrieval tasks leads to better retention than just using generative

tasks and better comprehension than using only retrieval tasks, which fits with the logic of the Feynman Technique especially well (Obergassel et al., 2025; Roelle et al., 2023).

Free Recall

Free Recall is a type of retrieval practice that encourages students to develop a better memory. It allows learners to recall the learned information and store it in the long-term memory without using notes or books. Students need to try to remember the information being learned instead of re-reading it. This method is ideal because it enhances the delayed learning and memory retention of the learned information (Rea et al., 2022; Serra et al., 2025). Studies have shown that retrieval practice is more effective than the use of restudy and rereading in enhancing student learning (Aljabri, 2024; Bego et al., 2024; Klier & Buratto, 2023)

Repeated retrieval can also facilitate recognition, conceptual retention, and transfer when practice is repeated sufficiently and tested after a delay (Corral & Carpenter, 2025; Huang & Cao, 2026). These results indicate that free recall is an effective cognitive learning strategy for promoting the students' independent learning, although it depends on the task's demands and the way it is implemented (Lima & Buratto, 2024; Redifer et al., 2025).

Adaptive Flashcards

Adaptive flashcards are an effective learning tool because they incorporate active recall and personalization of reviews. Unlike passive learning activities such as rereading notes, flashcards require learners to generate answers before checking them, which promotes memory consolidation and concept mastery (Huang et al., 2025; Jape et al., 2022). Studies on vocabulary acquisition demonstrate that using flashcards can enhance the recall performance and learning outcome, while digital flashcards can surpass paper and word lists in structured learning contexts (Le & Luong, 2023; Xodabande et al., 2022). Their effectiveness can be further enhanced by implementing spaced repetition, which enhances long-term retention and facilitates knowledge transfer compared to non-spaced reviews or lectures (Price et al., 2024; Santhosh et al., 2024).

Modern adaptive systems go further by modifying the review interval based on the learner's performance and the item's difficulty, thus focusing on the most challenging concepts and reducing the redundancy of reviewing well-learned items (Huang, 2025; Jape et al., 2022; Xiao & Wang, 2024).

Adaptive Assessment

The advent of computerized adaptive testing as a modern innovation in the field of education has led to the gradual replacement of some of the standardized assessment models that were in wide use in the past. Unlike the

traditional standardized tests used in the past, computerized adaptive assessments utilize artificial intelligence to generate tests that best suit the needs of individual students based on their responses to initial items (Zhuang et al., 2024).

In addition, AI-based quiz generation was found to facilitate the production of dynamic assessments, which are able to adapt the difficulty of questions and provide a more personalized experience (Al-Gburi, 2025). By doing so, computerized testing systems aim to enhance the efficiency of assessment and offer a more accurate representation of a student's knowledge and progress in learning.

When used together, the reviewed strategies have proven to be more efficient than a passive review. Gamification offers students an incentive to actively participate in the game, which motivates them to learn and engage with the material. The active recall and retrieval practice allow students to better remember the information learned in the game. The spaced repetition and adaptive flashcards focus on the concepts that students have not learned yet, rather than what they already know. The adaptive assessment feature is individualized to match a student's ability. LearnSync AI can be used as an all-in-one platform offering each of these four elements to help students with their self-directed learning journeys.

Performance Evaluation of Language Models and Inference Systems

The performance of large language models and their inference systems is a critical consideration in enabling AI-driven educational platforms to deliver fast, reliable, and responsive learning experiences. In digital learning environments, delays in processing user inputs and generating responsive outputs could adversely impact student engagement and learning continuity. Therefore, contemporary LLM systems are engineered with optimized inference and serving architectures that improve their ability to handle multiple user requests efficiently. Research on distributed model serving systems reveals that optimized inference techniques could enhance system throughput and response efficiency in transformer-based language models (Yu et al., 2022).

In addition, research on memory-efficient LLM serving shows that optimized resource management and computational power contribute to reduced latency and enhanced real-time performance of AI service processes in large-scale settings (Kwon et al., 2023). The two qualities are significant for the educational field that implements generative AI to guarantee effective performance and continuous support for its participants.

LLM in Materials and Context

Large language models (LLMs) are increasingly playing an important role in assisting students to interact with complex and lengthy learning materials. The initial transformer LLMs demonstrated remarkable skills in understanding and producing semantically meaningful text through learning semantic, syntactic, and

contextual relationships between language elements (Shi et al., 2024; Yan et al., 2023). As such models continued to evolve, they exhibited enhanced capabilities of in-context learning, instruction following, and reasoning, which broadened their application potential in various aspects of education as reading, writing, and explanation assistants (Shi et al., 2024; Wang et al., 2024). While the early forms of such models had relatively small context windows, recent advances in long-context modeling have enabled the LLMs to operate with considerably larger input sizes, thus improving their capacity for maintaining coherent conversations, topics, and contexts (Chen et al., 2023; Xu et al., 2023; Yen et al., 2024).

In educational settings, these improvements enable more personalized explanations, adaptive pacing, real-time feedback, and better synthesis of dense materials during independent study (Sharma et al., 2025; Wang et al., 2024; Zhui et al., 2024). However, the evidence also shows that effective support depends not only on larger context windows, but on retrieval, context management, and careful instructional design, given persistent risks related to hallucinations, bias, privacy, and student overreliance (Mei et al., 2025; Xu et al., 2023; Yan et al., 2023).

Real Time Feedback

Large language model–based systems provide an important source of immediate formative feedback within digital learning environments. In

conventional systems, delays in response generation can hinder students' focus during active instructional activities such as retrieval practice and review. By contrast, AI-based systems are capable of providing near real-time feedback and guidance that promotes student engagement and persistence in learning activities (Kasneci et al., 2023).

Reducing the response delay also enhances the interaction quality between the user and the computer, thus creating a lesser distraction to the learner. It enables the learners to detect errors and amend them promptly, thus improving the performance of the task, engagement, and behaviors such as revision (Meyer et al., 2024).

Model Specialization and Dynamic Routing

Usage of multiple large language models within one system is vital to the system's efficiency and quality. Instead of building a system around one model, which can lead to unnecessary complications, engineers can utilize multiple LLMs and use each for the types of queries it is better suited for. Routing methods for LLMs suggest that queries can be divided and sent to different language models based on their complexity and requirements for the most optimal results (Ong et al., 2024).

In this approach, less resource-intensive processes are farmed out to less capable but more efficient models, while more involved requests are delegated to larger ones. This routing mechanism allows for a better utilization of computational power, as well as faster response times, resulting in an improved experience when using the application in real time. The performance of an AI-based platform is determined by more factors than just the language model it uses. Fast processing, responsiveness, contextual awareness, and an optimal division of labor between different aspects of the AI are all important considerations. LearnSync AI is designed in such a way as to allow for all of these to be achieved at once, providing users with quick answers, processing long documents with ease and providing immediate feedback.

Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic model utilized in intelligent tutoring systems for tracking students' knowledge of a given skill based on their prior responses, including both correct and incorrect answers. This model enables the calculation of the probability that a student has learned a particular concept, thereby facilitating the tracking of knowledge acquisition in a dynamic context. The Bayesian knowledge tracing model has been widely adopted in various educational systems because of its relative simplicity in

application, enhanced interpretability, and reduced computational demands compared to alternative approaches (Bulut et al., 2023).

Tracing Models: BKT, DKT, and PFA

Bayesian Knowledge Tracing, Performance Factors Analysis, and Deep Knowledge Tracing are three different models that perform similarly in several aspects of student learning prediction. First, while BKT is based on an underlying process of student knowledge states' transitions in a Hidden Markov framework, representing latent knowledge states as binary values updated over time, this model is more intuitive for sequential learning and provides more interpretable results (Bai et al., 2024; Bulut et al., 2023). In turn, PFA is a logistic alternative to conventional knowledge tracing models, primarily distinguished by its ability to predict future successes and failures based on past successes and failures, learning each knowledge component as a function of performance on items linked to multiple skills, and being used occasionally in some of the latest studies (Lee et al., 2025; Zhang et al., 2025). Finally, DKT is a recurrent model applied to sequences of exercise responses and used extensively in the literature due to its ability to capture complex student response patterns, with later modifications like AT-DKT demonstrating high AUC and accuracy improvements across multiple datasets (Lee et al., 2025; Liu et al., 2023).

However, the literature consistently highlights a trade-off between accuracy and interpretability: deep knowledge tracing models can capture

complex behaviors effectively, especially when a large training dataset is available, but they can also be opaque and require extensive tuning, while BKT and other transparent models are generally easier to interpret in an educational context (Bai et al., 2024; Bulut et al., 2023; Zhou et al., 2024). While more complex models are important, recent studies suggest that simpler models also have practical value since BKT still enables personalized intervention through interpretable probabilistic parameters, and recent evidence suggests that PFA can still be competitive while BKT can be more stable than more complex models against concept drift (Bulut et al., 2023; Lee et al., 2025). Overall, recent research suggests that choosing between BKT, PFA, and DKT involves balancing predictive power, transparency, scalability, and robustness rather than finding a universal model for all education contexts (Abdelrahman et al., 2023; Bai et al., 2024; Zhou et al., 2024).

BKT Selection

Among the described tracing knowledge models, including Performance Factors Analysis (PFA) and Deep Knowledge Tracing (DKT), the researchers have chosen Bayesian Knowledge Tracing (BKT) to implement in the system. The choice of this method is justified by the possibility of applying it in adaptive learning settings and the relative simplicity of the model compared to the alternatives.

Unlike DKT that requires big datasets and high performance computation, and PFA that is focused on the counts of performance, BKT provides a lightweight framework that keeps track of the learners' responses individually. This model can be easily interpreted, thus allowing a better understanding of the learning process. For this reason, BKT was chosen to address the system's requirement of continuous temporal tracking of performance in an adaptive flashcard or assessment system.

Bayesian Knowledge tracing provides an updated estimation of knowledge for each student after each response, allowing for occasional guesswork and lapses in concentration, so that neither unduly influences the estimate. It has a light enough footprint to run in real-time, and the resulting estimates can be readily explained in simple terms. LearnSync AI makes use of these estimates to set the difficulty level for Adaptive Flashcards and the Adaptive Test Mode, to let each student practice at their own pace.

SIGNIFICANCE OF THE STUDY

This study has resulted with the development of LearnSync AI, a personalized learning platform which reinforces independent learning through adaptive activities, AI-assisted study strategies, gamified review, and learning analytics. The developed system benefited the following:

Pasig Christian Academy. The system provided the school with a central, AI-assisted learning platform to complement their current independent

learning process, with the integration of adaptive learning, personal assessment, and verified instructional material in one secure environment. Since the system is designed to be asynchronous and for home use, it would enrich the school's curriculum without interrupting their current operational structure.

Educators. The system provided the educators with a secure portal for uploading and managing the official resources that LearnSync AI uses. With the restriction of the AI-generated assistance based on teacher resources only, curriculum and academic integrity are prioritized. The educator dashboard also provides an overview of the students' progress and knowledge statistics without increasing their workload.

Junior High School Students. The system provided the students with a study companion that offers activities suited to their level. The combination of the system's adaptive practice, AI-assisted guidance, gamified review, and personal analytics in one platform is designed to encourage the students to engage in independent learning while monitoring their progress for their Senior High School preparations.

Researchers. The study provided the researchers with an opportunity to develop their knowledge and skills in regards with software development, educational technology, artificial intelligence, adaptive learning, and Bayesian Knowledge Tracing.

Future Researchers. The study may be used as a reference for further studies regarding AI-powered educational platforms, and as a basis for future

studies with regards to adaptive learning, Generative AI, Bayesian Knowledge Tracing, and personalized learning technologies.

STATEMENT OF THE PROBLEM

As students in Junior High School learn more difficult subjects, independent learning becomes necessary to supplement what they are taught at school so that they can be prepared and accustomed to the Senior High School curriculum. Students at Pasig Christian Academy do this through independent learning with materials provided by the instructors and the required textbooks of their subjects. However, the current practice is not very personalized, adaptive, or monitored. In order to supplement the students' independent study, the researchers have designed LearnSync AI, a program that uses Generative Artificial Intelligence and Bayesian Knowledge Tracing to personalize, adapt, and monitor the students' independent learning. Although the students' independent learning covers all of their subjects, the implementation and evaluation of LearnSync AI in this study were done using Science as the learning content.

This study therefore evaluated the accuracy of the program's implementation, the features developed to address the identified needs, the learning progress made as monitored by the Bayesian Knowledge Tracing component, and the overall acceptability of the product as addressed by the following questions:

1. What are the current independent learning practices and challenges of the Junior High School students of Pasig Christian Academy?

2. What features and functionalities, including the integration of Generative Artificial Intelligence and Bayesian Knowledge Tracing, can be incorporated into the system to address the identified challenges and requirements?

3. How can the correctness of the implemented Bayesian Knowledge Tracing engine be verified against manually computed knowledge estimates?

4. What change was observed in the students' Bayesian Knowledge Tracing (BKT) knowledge estimates recorded by LearnSync AI across their practice opportunities during the implementation period?

5. What is the level of acceptability of LearnSync AI as evaluated by the intended users, specifically Junior High School students, the educator, and information technology practitioners, in terms of the following eight ISO/IEC 25010 software quality characteristics:

    - Functional Suitability

- Performance Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

SCOPE AND LIMITATIONS

This study focused on the design, development, implementation, and evaluation of LearnSync AI, an independent learning platform that supports Junior High School students with adaptive learning, Generative Artificial Intelligence (Generative AI), Bayesian Knowledge Tracing (BKT), gamification, and learning analytics, all in one platform.

Scope of the Study

The platform supports the learning process of the school's current learning process and is not designed to supplant the learning that takes place in classrooms during instructional time, but rather is meant as a supplement using approved material from educators.

An educator module enables authorized educators to upload their resources and track student learning and engagement statistics. The uploaded contents on this platform form the sole basis of information for every artificial intelligence initiative, ensuring compliance with the syllabus.

LearnSync AI uses a classical four-parameter Bayesian Knowledge Tracing (BKT) model to estimate each learner's knowledge from their performance in Adaptive Flashcards and Adaptive Test Mode. These estimates personalize the learning activities: the selected questions and the difficulty of the questions in AI-powered games, including Crossword, Term Matching, and Timed Trivia.

Generative AI promotes independent learning by providing AI-assisted study methods such as the Feynman Technique and the Free Recall, personalized feedback, and an AI-learning assistant that answers questions based solely off of educator-uploaded materials.

The study evaluates LearnSync AI using the ISO/IEC 25010 Software Quality Model such as Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

The implementation and evaluation engaged 49 Junior High School students from Grades 7 to 10 of Pasig Christian Academy during the Academic Year 2026 - 2027 who comprised for 50 percent of the 98 students in the accessible population, one educator who took charge of the learning materials, and 10 information technology practitioners who assessed the tool from the

technical perspective. Although the respondents belonged to four grade levels, they were regarded as a single study population, and comparisons between grades were beyond the purview of the study. Instead, the evaluation explicated the students' learning progression via the BKT knowledge estimates generated by the platform throughout the practice opportunities.

Limitations of the Study:

The implementation lasted three weeks, one of which was spent using the product. Out of 49 chosen students, 46 opened the platform, and three did not, so there is no recorded evaluation. Three weeks is a short period in the context of a school curriculum, and thus the study only observed students' progress in the short term rather than assessing their overall performance in the long run.

The platform is conducive to concept-based learning and review. Specialized Mathematics, that is, symbolic computation, equation solving, graphing, step-by-step problem solving, is outside its scope, and so where Mathematics was a given grade level's trickiest subject, the study used another subject that the grade level identified and that lends itself to concept review.

Knowledge estimates come only from Adaptive Flashcards and Adaptive Test Mode; describe performance within LearnSync AI; and are never compared against school examinations or grades. The games, the Feynman Technique, and Free Recall either read the estimate or give feedback without changing it; thus, a student who used these features mostly produced little knowledge data,

and the records tracked the use of assessment modules rather than the whole platform. This was intentional. The BKT model only updates on answers that are either correct or incorrect, while Free Recall and the Feynman Technique give a score from the AI that may change slightly even when the same answer is submitted again. Adding these scores would bring the AI's grading errors into the knowledge estimate, which the model's guess and slip values were not designed to account for. Flashcard ratings, on the other hand, are the student's own report of recall and are not checked against an answer, so part of each estimate rests on self-assessment. The classical four-parameter BKT model applies a learning transition after every opportunity for practice; thus, estimates tend to rise with practice. A rising estimate is thus not by itself evidence of learning; the study read the change across successive practice opportunities and the share of concepts reaching the 85 percent threshold (Table 21 and Figure 19) instead. Since the model holds no prior evidence about an individual, every student begins with the same default probability; thus, the estimates become personal only as practice accumulates.

There was no control group, so the recorded progress cannot be credited to LearnSync AI alone; classroom instruction, study outside the platform, prior knowledge, and other resources may also have contributed.

Every AI feature is based solely off of educator uploaded resources and students cannot upload material therefore these resources are only as good as what the educator provides. The platform requires internet connection and will accept plain-text PDFs up to 20 MB and image or scanned documents, up to 20

pages, read through Optical Character Recognition which is dependent on the quality of the page which may be blurry, low resolution, poorly lit, skewed which reduces the quality of AI generated support. The platform does not offer offline capabilities, handwritten text recognition, voice interaction, payment processing, automated billing or enterprise level server scalability.

The platform relies on third-party services: Gemini API for AI generation and evaluation, and Supabase for storage and semantic retrieval. Preliminary tests have found slow responses to be the result of latency in the Gemini API, while the database and hosting services have performed normally. Responsiveness therefore relies on providers beyond the researchers' control, and generation may fail when a provider is unreachable or rate limits are exceeded.

CHAPTER II

INTRODUCTION

The chapter will discuss the methodology of designing, developing and evaluating LearnSync AI, a web-based platform designed to aid in the independent learning of the Junior High School students, equipped with the Educator Portal for teachers to manage the learning materials that should be accessed by the students. The following sections will present the technical design and development, algorithms and data structures of the system.

The application was designed with two types of users in mind. The educator uploads and manages learning materials to ensure that all information has been verified and validated by a qualified teacher and views students' progress through an analytics dashboard. On the other hand, the student interacts with the material and artificial intelligence (AI)-driven tools like Adaptive Flashcards, Adaptive Test Mode, Feynman, Free Recall, AI games, and an analytics dashboard. BKT is used to track knowledge traces and personalize learning activities based on individual performance needs. The chapter concludes by providing an implementation and evaluation of the application. First, the platform was implemented at Pasig Christian Academy for three weeks. Besides, an instrument that conforms to the ISO/IEC 25010 quality model was utilized to evaluate the application.

RESEARCH DESIGN

This study adopted a developmental research design, which is quantitative. In education technology, developmental research is characterized by the creation and systematic evaluation of an instructional tool or media to ensure it is maximally efficient and stable (Angeles & Cortez, 2023). Thus, it helped transition the software from design to an actual product tested by the target audience and implemented by the researchers, which helped meet the specific needs of the project participants while relying on quantified data.

The project was conceptualized into three phases. The first one was concerned with the analysis and design, which included the identification of learning problems of the Junior High School students and the supervisory requirements by the educator, to inform the creation of the Student and Educator Portal layouts, the study space architecture, the BKT routing rules, and the PlayMe gamification mechanics. The second comprised the development, which included the configuration of the Supabase relational database with the Row-Level Security policies, the programming of the generative AI parser, the implementation of the real-time BKT scoring engine, and the construction of the educator analytics dashboard. The last one was the evaluation, which was performed by deploying the solution over a span of three weeks at Pasig Christian Academy, where the educator utilized the platform to upload materials, while the students engaged with the tool independently.

Two methods of collecting quantitative data were applied. The first one included extracting the system's logs of students' interaction with the Adaptive Flashcards and Adaptive Test Mode, where BKT algorithm calculates the probabilities of students' knowledge states based on their responses. Another method involved conducting an assessment of the software's quality based on ISO/IEC 25010 standard. Within the scope of this assessment, participants filled out a questionnaire, which included a four-point Likert scale. The results of the analysis were evaluated using descriptive statistics to define the level of acceptance of each of the eight characteristics.

ARCHITECTURAL FRAMEWORK

LearnSync AI is designed based on a two-portal, three-tier system. The Educator Portal is responsible for uploading learning material, restricting the content according to the grade level, and monitoring student knowledge. The Student Portal is used for independent study, adaptive assessment, and gamified reviews. The data flows through the Presentation, Application, and Data Tiers when the user submits any request, and the response comes back through these tiers.

1. **Presentation Tier -** is the system's interface layer, which consists of 4 areas. The Educator Portal is intended for uploading text files in PDF format or image files of printed material, designating what grade levels can

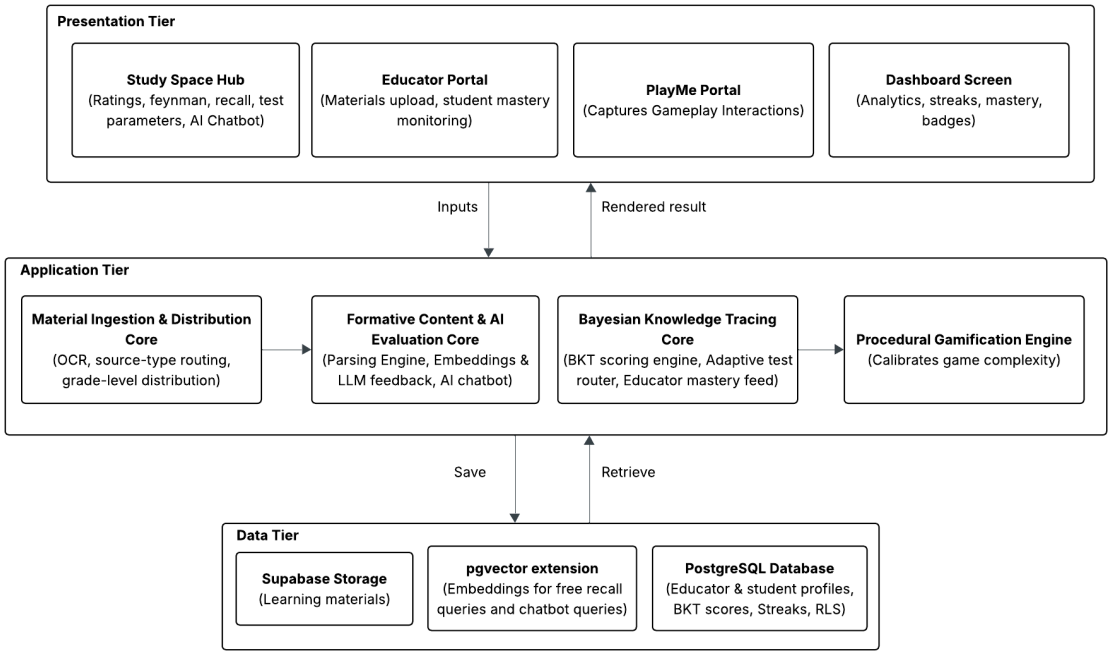
see this material, and evaluating the students' knowledge in real-time. The Study Space Hub is where students open the uploaded material, rate the flashcards generated by the system, perform the Feynman and Free Recall exercises, set up practice tests, and communicate with the AI chatbot. The PlayMe Portal is where students play the review games, and the Dashboard Screen is the screen where users can monitor their progress, see their streaks and knowledge curves, and collect achievements.

2. **Application Tier -** is the system's processing layer that has 4 components. The Material Ingestion and Distribution Core is responsible for distributing the uploaded material depending on the source, with plain-text PDFs going directly to the parsing engine and image files needing OCR before entering the distribution loop. The Formative Content and AI Evaluation Core is where the system extracts the required terms and evaluates the Free Recall and Feynman Responses using vector embeddings and LLMs. The BKT Core is where the flashcard ratings and test responses are processed into the knowledge model, where the procedural tests are tuned in difficulty, and where the data is processed for the dashboards. The Procedural Gamification Engine is where the knowledge percentage informs the game difficulty.

3. **Data Tier -** is the storage layer of the system, which consists of three main elements: the Supabase Storage where the uploaded files are stored and the pgvector extension that stores the vector representations for the similarity search used in Free Recall and chatbot features. The PostgreSQL database stores the educators' and students' accounts, the grade levels, BKT knowledge scores, streaks, and other RLS-controlled data.

Figure 1.

Architectural Framework



Note. The figure shows the dual-portal, three-tier architecture of LearnSync AI and the flow of data between the Presentation, Application, and Data Tiers

ALGORITHM

This section provides the mathematical foundation of the cognitive modeling core of LearnSync AI. By implementing the classical four-parameter Bayesian Knowledge Tracing (BKT) model, the platform relies on a probabilistic learner model that is still considered one of the most prominent and interpretable methods for knowledge tracing and is actively used in current studies (Bulut et al., 2023). It is a dynamic hidden Markov model with observed nodes that track the latent knowledge state of a student related to a particular concept through the set of responses made by the student without substantial server loading (Abdelrahman et al., 2023).

1. **BKT Mathematical Model -** BKT estimates the probability that a student has learned a concept from their chronological sequence of correct and incorrect answers. Four interpretable parameters govern the model (Bulut et al., 2023): the initial knowledge state $P(L_0)$, the prior probability that a student already knows a concept; the transition probability P(T), the learning rate at which a student moves from not knowing to knowing; the guess probability P(G), the chance of answering correctly without knowing; and the slip probability P(S), the chance of answering incorrectly despite knowing.

   a. **Initial Knowledge State** $P(L_0)$: The prior probability that a student already knows or has learned a concept before interacting with any formative study modes or assessments on the platform.

b. **Transition Probability** $P(T)$: The learning rate represents the probability that a student shifts from an unlearned (not-known) state to a learned (known) state after a learning opportunity.

c. **Guess Probability** $P(G)$: The probability that the student has not learned the concept but is still able to answer correctly.

d. **Slip Probability** $P(S)$: The probability that a student who has learned the concept still answers incorrectly due to carelessness, cognitive fatigue, or a reading error.

As the student interacts with the application, their performance at any given learning opportunity *t* is captured as an observable binary variable (obs_t), where an answer is marked as correct ($obs_t = 1$) or incorrect ($obs_t = 0$). When a new observation is captured, the system executes a two-phase Bayesian update. First, the algorithm applies the evidence update phase to calculate the conditional probability of knowledge ($P(L_t \mid obs_t)$ based on the observation:

Formula when the students attempt is correct ($obs_t = 1$) :

$$\textbf{EQ (1):}\ P(L_t \mid obs_t = 1) = \frac{P(L_t)\,(1-P(S))}{P(L_t)\,(1-P(S)) + (1-P(L_t))\,P(G)}$$

Formula when the students attempt is incorrect ($obs_t = 0$) :

$$\textbf{EQ (2):}\ P(L_t \mid obs_t = 0) = \frac{P(L_t)\,P(S)}{P(L_t)\,P(S) + (1-P(L_t))\,(1-P(G))}$$

Second, the algorithm performs the transition update phase. This step calculates the projected probability of knowledge for the subsequent learning opportunity $P(L_{t+1})$ by factoring in the learning rate $P(T)$ as the student moves to the next card or question, assuming immediate formative feedback is processed:

$$\textbf{EQ (3):}\ P(L_{t+1}) = P(L_t \mid obs_t) + (1 - P(L_t \mid obs_t))\, P(T)$$

1.1. **Default Parameter Initialization** - LearnSync AI is newly developed and has no student interaction data, so it cannot calculate parameters from responses. Expectation Maximization, the standard method, requires a large dataset of prior responses (Bulut et al., 2023; Šarić-Grgić et al., 2024). The system therefore uses fixed values, which does not prevent it from working since tracking begins at a student's first interaction.

The knowledge tracing literature does not recommend specific numbers for these parameters, since published values are calculated from each study's own data. What it provides instead are limits on acceptable values. These prevent model degeneracy, a condition in which the model behaves illogically, such as lowering a student's knowledge after a correct answer. A recent systematic review documents three sets of limits still in use (Šarić-Grgić et al., 2024). As reported there, Corbett and Anderson proposed $P(G) < 0.30$ and $P(S) < 0.10$; Beck and Chang proposed $P(G) + P(S) < 1$; and Baker, Corbett, and Aleven proposed $P(G) < 0.50$ and $P(S) < 0.50$ (all as cited in Šarić-Grgić et al., 2024). The values used in this study were selected since they fall inside the accepted limits.

The following default parameter values were adopted for all freshly created concept nodes when a document was uploaded:

a. $P(L_0) = 0.10$. The initial amount reflects a 10 percent chance that a student already knows a concept at the time tracking begins. This conservative value was selected because the platform has no access to prior-knowledge data and does not perform a diagnostic assessment before the tracking begins, leaving no basis for a higher starting point. A low starting point requires knowledge to be built through subsequent recorded performance. Of the four parameters, $P(L_0)$ has the shortest-lasting impact: Each response multiplies the previous value, dampening the influence of the starting point after just a few responses.

b. $P(T) = 0.20$. The learning rate provides a student with a 20 percent chance of progressing from the unlearned to the learned state at each opportunity. The rate appropriate to a type of opportunity depends on what the opportunity comprises, and published research offers differing values for different kinds of activity (Badrinath et al., 2021; Pardos et al., 2013, both as cited Šarić-Grgić et al., 2024). Conventionally, values are derived from intelligent tutoring systems, where an opportunity takes the form of a complete guided problem with instructional feedback. In LearnSync AI an opportunity is a flashcard review or single test item which does not include feedback and where the student bears the full self-monitoring burden (Luo & Zhou, 2024) and so a moderately conservative value was adopted. This

investigation acknowledged that equal application of a single P(T) to these kinds of activity is a simplification.

c. $P(G)$ = Context-Sensitive (Dynamic). In standard BKT the guess probability is a single value per concept; LearnSync AI varies it by assessment mode, question format, and self-reported confidence. This is in line with established practice in that allowing guess and slip to vary between responses produces more accurate student models than using per-concept values, and question difficulty is one of the most studied extensions of this kind (Baker, Corbett, and Aleven, 2008, as cited in Šarić-Grgić et al., 2024). Varying P(G) is mathematically legitimate since the update formulas take any value between 0 and 1 for each response. In Adaptive Test Mode the values are not estimated but calculated: for each question format, P(G) is the chance of randomly guessing the correct answer, which depends on the number of answer choices.

In Adaptive Flashcard Mode, the value is based upon the student's confidence rating. Easy is set as P(G_easy) = 0.05, which represents high confidence that the answer was from their actual knowledge. Medium has P(G_medium) = 0.25, representing the possibility that some answers could be random guesses. Hard has P(G_hard) = 0.10, and will be logged as an incorrect response; since P(G) is reflected as (1 − P(G)) in the denominator of the incorrect answer formula, the lower value causes it to have a sharper decrease, so 0.10 ensures that a self-reported failure of recall decreases their knowledge appreciably.

In Adaptive Test Mode, the guess probability is based on the mathematical probability of getting the correct response by random guessing alone for that question type. Since the base chance of getting a correct answer by random chance is determined by the number of possible choices, these numbers are derived from the question type. The following P(G) values were adopted for each question format:

1. Multiple Choice (4 options): $P(G_mc) = 0.25$. With four answer choices and one correct answer, the likelihood of guessing correctly is 1/4 = 0.25.
2. True or False: $P(G_tf) = 0.50$. With only two possible answers, the likelihood of guessing correctly is 1/2 = 0.50. This increased guess rate means that correct answers on true/false items will contribute to a much smaller increase in knowledge value than other question types, due to their increased guess rate.
3. Identification: $P(G_id) = 0.05$. Open-ended identification questions require the student to type in the term, without any multiple choice answers. The chances of guessing the correct answer is incredibly low.
4. Fill in the Blank: $P(G_fib) = 0.05$. Similar to identification items, fill-in-the-blank questions require free recall of specific terms or phrases without cues. The absence of selectable options makes random guessing highly unlikely, justifying $P(G) = 0.05$.
5. Matching Type (4 terms, 6 definitions): $P(G_match) = 0.17$. For any individual term, the probability of selecting the correct definition from the

pool of six through random guessing is 1/6 ≈ 0.17, applied per matched item.

The rule preventing this is $P(G) + P(S) < 1$, proposed by Beck and Chang (2007, as cited in Šarić-Grgić et al., 2024) and documented as still in use by the current systematic review of the field.

Every combination used here passes. With $P(S) = 0.15$ throughout, Identification, Fill in the Blank, and Easy-rated flashcards use $P(G) = 0.05$, totalling 0.20; Hard-rated flashcards use 0.10, totalling 0.25; Matching Type uses 0.17, totalling 0.32; Multiple Choice and Medium-rated flashcards use 0.25, totalling 0.40; and True or False uses 0.50, totalling 0.65. Every total is below 1.

Seven of the eight formats also pass both the stricter $P(G) < 0.30$ of Corbett and Anderson and the looser $P(G) < 0.50$ of Baker, Corbett, and Aleven (2008; both as cited in Šarić-Grgić et al., 2024). True or False is the only exception, which cannot be avoided with a two-choice format where random guessing succeeds half the time. The practical result is that a correct True or False answer raises knowledge far less than other formats, correctly reflecting how little it proves.

d. $P(S) = 0.15$. A student who knows a concept still answers incorrectly 15 percent of the time through carelessness, misreading, or fatigue. This value passes the degeneracy rule for every format and the P(S) < 0.50 limit of Baker, Corbett, and Aleven. It exceeds only the strictest limit, P(S) < 0.10 of Corbett and Anderson (1995, as cited in Šarić-Grgić et al., 2024),

which was set for tutoring systems with real-time supervision and immediate correction. LearnSync AI is self-directed, where students work alone and are more prone to lapses in attention (Luo & Zhou, 2024).

The model comprises two classes of values. P(L), the knowledge estimate, is the value that varies with every response. The four other parameters are static; they specify how each update should proceed and are unvaried during execution. The Bayesian mechanism adjusts the estimate rather than the parameters, which are thus characterized here as selected configuration values constrained by published limits, rather than values toward which the system converges. Fixed parameters are typical for BKT implementations that do not have access to historical training data, and the literature favors computing them from data when sufficient data is available (Šarić-Grgić et al., 2024). Recomputing the parameters based on collected interaction data after deployment is indicated as future work.

1.2. **Step-by-Step Computation of Student Knowledge** - the following trace shows three consecutive multiple-choice attempts with $P(L_0) = 0.10$, $P(T) = 0.20$, $P(S) = 0.15$, $P(G) = 0.25$. Each attempt first applies the evidence update followed by the transition update.

Attempt 1 - Correct $obs_t = 1$, prior $P(L_0) = 0.10$:

$$P(L_0 \mid obs_t = 1) = \frac{(0.10)(1-0.15)}{(0.10)(1-0.15)+(1-0.10)(0.25)} = 0.27$$

$$P(L_{0+1}) = 0.27 + (1 - 0.27)\,0.20 = 0.42$$

Attempt 2 - Correct $obs_t = 1$, prior $P(L_1) = 0.42$:

$$P(L_1 \mid obs_t = 1) = \frac{(0.42)(1-0.15)}{(0.42)(1-0.15)+(1-0.42)(0.25)} = 0.71$$

$$P(L_{1+1}) = 0.71 + (1 - 0.71)\,0.20 = 0.77$$

Attempt 3 - Incorrect $obs_t = 0$, prior $P(L_2) = 0.77$:

$$P(L_2 \mid obs_t = 0) = \frac{(0.77)(0.15)}{(0.77)(0.15)+(1-0.77)(1-0.25)} = 0.40$$

$$P(L_{2+1}) = 0.40 + (1 - 0.40)\,0.20 = 0.52$$

Two consecutive correct answers raised knowledge from 0.10 to 0.77, while one incorrect answer lowered it to 0.52, a significant but moderate decrease that demonstrates the effect of the slip parameter that prevents drastic reductions in knowledge due to individual slips.

2. **Algorithmic Implementation in Flashcard Mode -** Within the study space, spaced repetition manages how the flashcards behave while BKT calculates the student's knowledge of a concept. Flashcards are generated from the educator's uploaded materials; students cannot write cards by hand. A student creates, renames, and deletes decks, but every card is generated against a selected material in one of two formats: the concept's term, which uses no artificial intelligence, or a question written by the large language model. In both, the back

of the card is the educator's stored definition reproduced verbatim, so the answer a student is assessed against is always teacher-approved content.

Opening a session loads the knowledge estimates for the deck's concepts. No record is made in advance; a concept that has yet to be determined is simply given the default prior $P(L_0) = 0.10$ on reading, and a stored record is not made until the first practice possibility. The student flips a card to see its definition and marks it Easy, Medium, or Hard.

Each rating corresponds to a binary observation (correct or incorrect response), a guess probability that reflects the student's reported confidence in their response, and a Leitner review interval (Durrani et al., 2024). Easy implies the student remembered the card: the system marks the response as correct $(obs = 1)$, applies the low guess probability $P(G_easy) = 0.05$ so that a confident recall is credited to knowledge rather than luck, giving the largest increase, and review the card in seven days. Medium implies the student remembered the card, but it was not easy: also a correct response $(obs = 1)$, but a higher $P(G_medium) = 0.25$ produces a smaller, more conservative boost, and the card will be reviewed in three days. Hard implies the student did not remember the card: the system marks the response as incorrect $(obs = 0)$, applies$P(G_hard) = 0.10$, and reviews the card the next day for immediate reinforcement. The increasing intervals ensure that the cards the student forgets are reviewed more often.

A rating modifies the knowledge estimate only when the card is due for review. First, the system examines the card's next review date: if the date has passed or is currently due, the rating is a valid practice opportunity, so the two-phase Bayesian update is performed, the evidence phase followed by the transition phase, the new $P(L_t + 1)$ saved, and the card rescheduled by its interval. If the card is not due, the rating is treated as extra practice. The student sees the card and receives feedback but no estimate or review date modification and is informed that their rating was not counted. This preserves the model's assumption that each observation is an independent test of latent knowledge; without this rule, a student could rate a card Hard, re-drill it seconds later having just read the answer, rate it Easy, and cause the estimate to rise most rapidly on the concepts they know least. A newly created card is due immediately, so the first pass through a deck always counts.

Review dates fall on midnight in the Philippine time zone rather than precisely twenty-four hours after rating, so a student who studies at eight in the evening finds the card waiting after school the next day

3. **Algorithmic Implementation in adaptive test mode** - Within the adaptive test mode, BKT provides the primary mechanism for adjusting the difficulty of the generated assessment sessions. Students select which type or types of questions (multiple choice, identification, matching type, fill in the blank, or true or false) but not the length of the test. The length is derived from the material: each

extracted concept receives exactly one question of the selected type(s), distributed so that matching type appears only in groups of four terms to six definitions. One question per concept is sufficient to provide one practice opportunity per concept, since repeated questions on the same concept within one session would not provide independent observations.

Each response is graded as a binary value (1 correct, 0 incorrect) by the system rather than by the language model, and may be revised for as long as the test remains open. The knowledge estimates update once, at submission. Every item is then applied in question order as a separate practice opportunity, each using the guess probability of its own format (0.25 for multiple choice, 0.50 for true or false, 0.05 for identification and for fill in the blank, and 0.17 applied per matched item in matching type) together with the fixed slip probability $P(S) = 0.15$, so the knowledge gained from a correct answer scales to how much chance the format allows. An unanswered question is graded incorrect and still applied. The results update the student's concept-specific history in the central database.

When the student requests a new test on the same topic, the system refers to that history to adjust the difficulty of the new items. The document's concept estimates are pooled into the document knowledge index M_{doc}, specifically the arithmetic mean of those probabilities, and that value is converted to a difficulty tier that modifies the prompt to the language model. At or above 0.85 the model is prompted to produce advanced-tier questions emphasizing analytical scenarios, synthesis, and edge cases. Between 0.50 and 0.85 it will

produce intermediate-tier questions focusing on conceptual comparisons, multi-concept recall, and practical applications. Below 0.50 it will produce simpler, more highly scaffolded retrieval questions focusing on definitions and direct conceptual associations.

Both the tier and the knowledge value that it came from are stored along with the generated test, so that the calibration of any previous session can be confirmed. Such session-based adaptation allows students to retain full control over their test setups, while allowing the system to adapt the content's cognitive challenge to each student's ability.

4. **Document knowledge and Gamification Calibration** - While BKT tracks individual concepts as unique, high-fidelity knowledge components the system needs to provide an aggregated progress index. While a document is uploaded the ingestion engine harvests N individual concept nodes and the application layer combine their independent probabilities via arithmetic mean. The overall knowledge rate Mdoc is then computed at any point in time based on the most recent running BKT figures:

$$\textbf{EQ (4):}\ M_{doc} = \frac{\Sigma\, P(L_{current,\,i})}{N}$$

Where M_{doc} is the percentage of the document that the student knows reflected as a progress bar on the dashboard; N is the number of unique concepts parsed from the document, and $P(L_{current},\ i)$ is the running BKT state

for the i-th concept under the student's profile. These are each derived by applying the evidence and transition equations in chronological sequence. For a concept with no recorded practice session the initial $P(L_0) = 0.10$ is used; each later attempt is computed using the binary result of that attempt, calculating the conditional probability according to the evidence equation and executing the transition update. The value used is the last-saved $P(L_t + 1)$ at the time of calculation, allowing the overview to respond to real-time progress.

The gamification module uses this score to determine the difficulty of generated games (crosswords, term matching, timed trivia) but not as a continual percentage. On initialization the system maps the current M_{doc} to one of three difficulty tiers (foundational for below 0.50, intermediate for 0.50 to 0.85, and advanced for 0.85 and above), selecting a corresponding rule set governing vocabulary complexity and clue difficulty along with a trivia timer of twenty, fifteen, or ten seconds respectively. A student at precisely 50 percent therefore receives intermediate content with a fifteen-second timer. Both the tier and underlying Mdoc values are saved with the game for later review.

The individual concept estimates also determine which concepts a game draws on. Concepts are selected by weighted random sampling in which a lower estimate carries a higher probability of selection, so the student practices what they are weakest at while mastered concepts remain reachable rather than excluded. For this selection alone a concept with no recorded estimate is weighted as though it stood at 0.50 rather than at $P(L_0) = 0.10$, so

never-practiced concepts rank below those the student is measurably struggling with and above those already mastered. Because selection is random rather than a fixed ranking, requesting a new game on the same material does not reproduce an identical one. This keeps gamified learning synchronized with cognitive progress while protecting students from frustration.

5. **Evaluation in Free Recall** - Within the study spaces, the Free Recall tab lets students type everything they remember about a lesson without looking at their notes. To grade this automatically the system compares the meaning of what the student wrote against the lesson's concepts, using Cosine Similarity over high-dimensional vector embeddings, in four steps.

First, the raw text is sent to the large language model, which extracts the discrete ideas the student discussed, producing one item per idea rather than one per paragraph. A point summarizing a whole paragraph would embed as an average of everything in it and would match no single concept closely.

Second, each extracted idea is embedded separately using the same model applied to the educator's material at upload. The lesson's concepts were embedded the same way during ingestion and are stored in the database through the Supabase pgvector extension

Third, each stored concept is compared against every one of the student's idea vectors and the highest comparison is taken as that concept's best match.

The comparison measures the angle between two vectors and is computed in the application layer over the vectors retrieved from the database:

$$\textbf{EQ (5):}\ Cosine\ Similarity\ = \frac{A \cdot B}{||A||\ ||B||}$$

where A is a stored concept vector, B a student idea vector, $A \cdot B$ their dot product, and $|A|$ and $|B|$ their magnitudes. A score near 1 means the student's idea points in the same directional meaning as the concept.

Fourth, a concept is designated Covered if its best match has a similarity that meets or exceeds the 0.70 threshold and Missed otherwise. The threshold is applied individually to each concept, and not to the submission as a whole. The reported score is the coverage ratio: concepts Covered divided by the total concepts in the lesson. The Missed concepts are then shown right away (the specific ideas the student did not mention) with written feedback. A coverage ratio (and not an average of raw similarity scores) keeps the number honest, as even unrelated vectors will score about 0.60 in a high-dimensional space and averaging would artificially boost every score.

The pgvector stores the embeddings and serves them to the application while the similarity itself is computed in the backend, as each submission compares every stored concept against every extracted idea (rather than doing one pairwise distance computation between the two fixed vectors).

6. **Feedback generation in Feynman Technique** - The Feynman Technique aids students in learning by explaining complex concepts in simple, accessible language as though teaching an inquisitive child aged 10. The application sends the requested explanation to a large language model as a single prompt and receives a single response containing three checks; the checks below describe what is evaluated, not three separate requests.

The Jargon Check scans the explanation against the glossary extracted during document upload. Complex terms used without being broken down, such as “photosynthesis” or “electromagnetism”, are flagged and returned to the student to explain more simply. The system then removes from that list the name of the concept the student was asked to explain, since naming it cannot itself be jargon.

Analogy Verification checks whether the student used an everyday comparison to make the idea clear, such as comparing an electrical circuit to water flowing through pipes. The system then rejects any comparison drawn from the lesson's own vocabulary, because comparing one idea from the subject to another from the same subject is an in-domain example rather than an everyday analogy and teaches a child nothing.

Finding the Gaps compares the explanation against the concept's stored definition to check whether anything is factually wrong, missing, or incomplete, producing an accuracy score from 0 to 100 that judges only whether what the student asserted is true, regardless of wording.

A second, independent simplicity score from 0 to 100 measures how well the explanation does for an audience of 10-year-olds, regardless of the accuracy; a hollow, circular explanation is thus penalized here instead of for incorrectness. Separating the axes ensures that a correct but jargon-filled explanation does not get penalized for its accuracy it did not have, and a simple but incorrect one will not be rewarded for its clarity.

The verdict of Correct, Needs Improvement, or Incorrect is taken by the system, from the accuracy score only: Correct at 80 and above, Needs Improvement from 30 to 79, Incorrect below 30. It is never derived from the model itself, because when prompted the model tends to let the simplicity axis drive its judgment, which would allow the badge to contradict the accuracy score it was given. Aside from the scores, the AI offers brief, upbeat feedback notes that highlight what the student did right, call out specific factual errors, and tell them how to improve in exactly those areas.

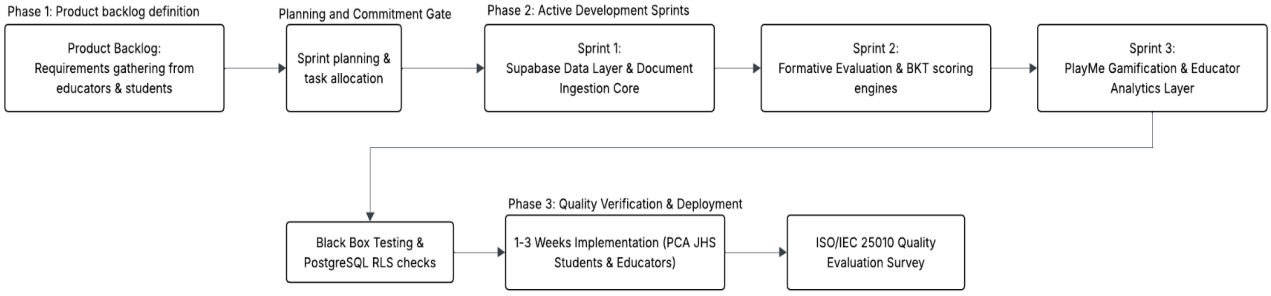
SOFTWARE ENGINEERING PARADIGM

The study was conducted using the Agile Scrum framework because the platform has multiple interrelated components that require constant testing and improvement. The framework allows for developers to create and improve the platform in repeated short cycles and make adjustments to meet the end-users' needs (Hodson et al., 2024). It also aids the detection of possible flaws at an

early stage and minimizes development risks, thus enhancing the quality of the final product (Strode et al., 2022).

Figure 2.

Agile SDLC



Note. The figure shows the Agile Scrum development cycle followed in building LearnSync AI, from product backlog definition to quality verification and deployment.

The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Product Backlog** - the researchers defined requirements from two different roles' perspectives. The educator uploads the material, chooses the grade levels that can see it, and tracks their progress via a class-level dashboard. The Student, in turn, studies the materials with four options (adaptive Flashcards, Free Recall with AI grading, typed explanations using the Feynman Technique, and an AI chatbot), creates personalized

practice tests in five different adaptive formats (based on their BKT) and a gamified review engine that scales its difficulty based on the same score.

2. **Planning and Commitment Stage** - the sprint planning created a breakdown of the backlog into modules, determined their complexity, set priorities, and created a timeline for the sprint. It helped to keep the sprints focused on the goal, prevent scope creep, and prioritize mathematical models and security measures over the actual coding of the software.

3. **Active Development Sprints** - The first sprint focused on creating the Supabase PostgreSQL tables for the educator and student profiles, grade level access control, materials and embeddings, Row-Level Security policies, and the LLM-based extraction code for concept, definition, and embedding parsing of the uploaded materials. The second sprint involved developing the four-parameter BKT calculations and the adaptive test routing algorithm, vector-based Free Recall assessment, the LLM prompt templates for the Feynman and chatbot, and the timestamps recording functions. Finally, the third sprint was used to code the backtracking crossword grid, trivia, and vocabulary matching algorithms, build the gamification and streak system, program the educator dashboard, and build the API backend and portal frontend.

4. **Quality Verification & Development** - the application was subjected to PostgreSQL RLS checks to verify the data isolation for both student-to-student and educator access boundaries and thus launched for a three week implementation at Pasig Christian Academy. Finally, an ISO/IEC 25010 quality evaluation survey (four-point Likert scale) was administered.

DATA FLOW DIAGRAM

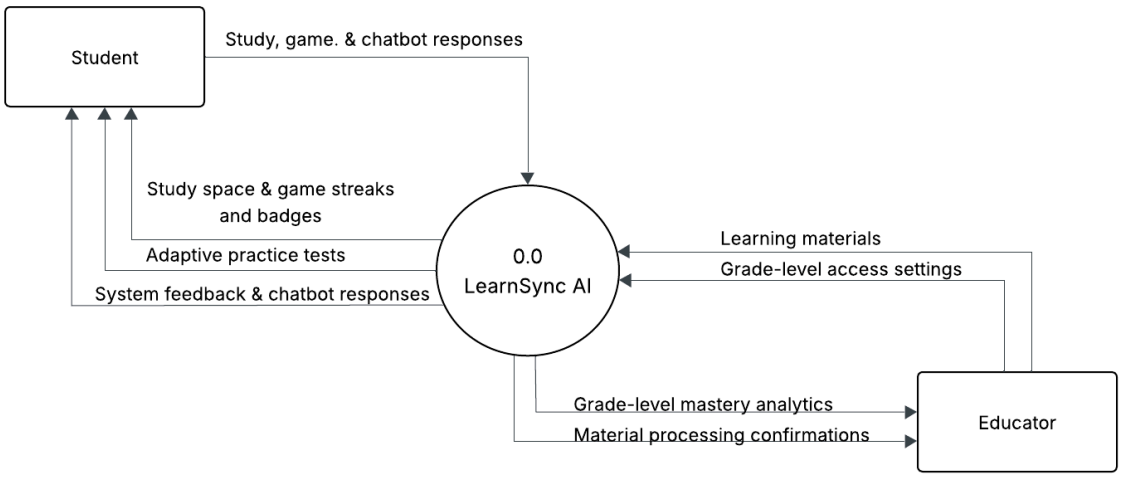
A Data Flow Diagram is used to depict different flows of information within the system. On the diagram below, Level 0 presents an overview of data flow, and it is possible to see how a teacher uploads the material and how the system provides tests, progress, and feedback to the students and educators. On Level 1, the process is separated into distinct activities and data stores.

1. **Level 0: Context Diagram -** the context diagram below represents the whole application as a single process, 0.0 LearnSync AI, with two external entities. The Educator external entity sends learning materials (plain-text PDF modules or images of printed pages) and grade-level access settings to the application and receives grade-level knowledge analytics and material processing confirmations from the application. The Student external entity sends study, game, and chatbot responses to the application and receives study space and game streaks and badges,

adaptive practice tests scaled to their knowledge level, system feedback, and chatbot responses from the application.

Figure 3.

Level 0 Context Diagram



Note. The figure shows LearnSync AI as a single process together with its two external entities, the Student and the Educator, and the data exchanged between them.

2. **Level 1: Data flow diagram** - The Level 1 diagram that was created, contains five processes, three data stores (D1: Study Spaces, Learning Materials and Embeddings; D2: Student Profiles and Knowledge Track; D3: Activity and Streak Logs), and two external entities.

    Process 1.0. Material Upload and Distribution Control, gets the materials and access parameters from Educator and saves the raw data and access rules in D1. Upon the Student opening their working partition it

interrogates D1 for matching access rules and returns the permitted materials. It also reads the aggregated knowledge data stream from D2 and feeds it to the Educator.

Process 2.0. Document Parsing and Semantic Ingestion, takes a raw file from D1, extracts keywords, performs an embedding of the text, saves it back to D1, and notifies the Educator that a file is being processed.

Process 3.0. Formative Study and AI Evaluation, takes care of flashcards, the Feynman Technique, Free Recall, and the chatbot. It gets responses and questions back from the Student, reads the parsed content back from D1, and gives feedback and chatbot answers. The AI generated flashcards are stored back into D1. For the knowledge tracking, it reads the current flashcard knowledge from D2, runs the BKT calculation, and writes the score back, logging each session to D3.

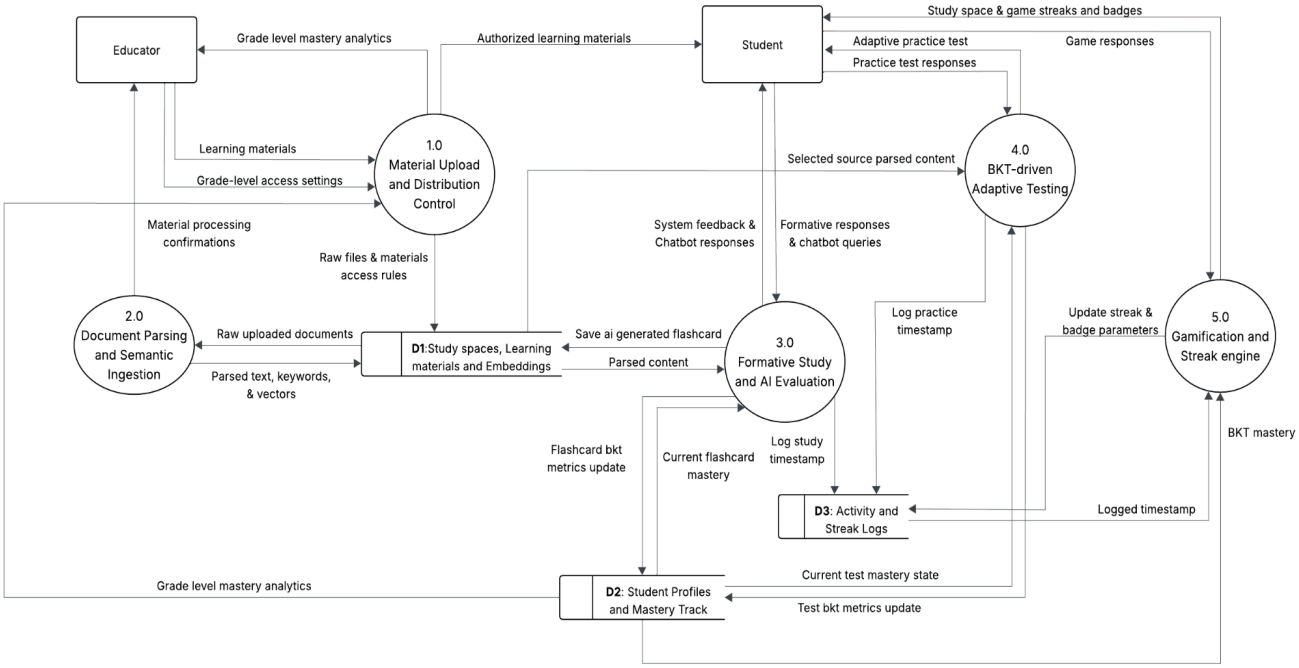
Process 4.0. BKT-driven Adaptive Testing, receives practice test responses from the Student, parses the content of D1 and reads the current test knowledge state from D2, applies BKT and returns tests that are adaptive at the student's level. Updated scores are written to D2 and session times logged to D3.

Process 5.0. Gamification and Streak Engine, receive game responses, read the document level score M_doc from D2 for calibration of the difficulty and the logged timestamps from D3 for the verification of

daily activities, update streak and badge records and return streaks and badges to the Student.

Figure 4.

Level 1 Main Process



Note. The figure shows the five processes, three data stores, and two external entities of LearnSync AI, decomposed from the single process in Figure 3.

TOOL TO TEST FOR ACCURACY

To ensure that the results are reliable and correct, the study made use of a multi-layered validation process that covered both the algorithm implementation as well as the functioning of the system itself. Since LearnSync AI makes use of no historical interaction data or model training, the evaluation did not make use of

many machine learning-specific metrics such as Confusion Matrix, Precision, Recall, F1-Score, or Mean Absolute Error. Instead, it made use of an alternative software verification method that sought to confirm that the BKT engine and Generative AI components follow the specified formula and behaviors.

1. **BKT Algorithm Implementation Verification**

   The BKT module was tested with deterministic algorithmic verification, which confirmed that it produces knowledge estimates equal to hand-computed values. The study does not test whether BKT tracks real-world knowledge, since it evaluates the skill with predefined parameters, not values learned by the algorithm of Expectation Maximization. The objective of the research was to ensure that the code does not have computational or logical errors in the calculation of the standard four-parameter equations.

   The researchers created test cases with an initial state, a sequence of binary answers, default parameters, the context-sensitive P(G) for the interaction mode, and the expected result that they calculated manually. The cases included different alternating sequences of correct and incorrect answers, consecutive correct, consecutive incorrect, and mixed answers, for all three flashcard levels and all five types of test questions. Each test was run in the program and compared to the expected result.

The correctness rate was measured as the Implementation Correctness Rate:

$$\textbf{EQ (6):}\ ICR = \frac{Number\ of\ Passed\ Test\ Cases}{Total\ Number\ of\ Test\ Cases}\ X\ 100$$

A case passed when the absolute difference was within ± 0.01 due to a floating-point variation. An indicator of a complete and correctly implemented model is an ICR of 100 percent, and any failure would be reported with an indication of the expected value, an actual value, and a deviation therefrom. It should be noted that such a verification checks the implementation's correctness, not a model's predictive validity (Bulut et al., 2023).

2. **Generative AI Functional Verification**

The Generative AI components were tested by checking the functionalities instead of resorting to model-based benchmarks as the study did not involve fine-tuning or retraining the model. The researchers uploaded a controlled learning resource, triggered each AI-powered functionality, and scored each outcome on four criteria. Contextual Relevance was used to score if the contents pertained to the topics and terminologies included in the uploaded resource. Factual Correctness measured the accuracy of the results compared to the source material. Difficulty Alignment ensured that the contents were appropriate for the

students' current level of knowledge to confirm the correct functioning of the adaptive pipeline with the parameter of knowledge. Content Consistency was used to ensure that repeated generations were not conflicting with each other, and although the phrasing varied, the facts, terminologies, and difficulty remained the same. The check was completed after the implementation period on the deployed site using an educator test account plus a new student test account outside the study cohort. The test included 37 outputs from the seven Generative AI components, all using pre-prepared inputs whose correct results were already known, and one repeated run for each component to test consistency. The outputs were produced and saved using an automated browser procedure, and each was rated pass or fail against the uploaded material. The results are reported in Chapter III.

3. **ISO/IEC 25010 Software Quality Model Survey**

To assess qualities from the end user perspective, the system was evaluated according to the eight characteristics listed above. An evaluation questionnaire was prepared based on these eight features and was distributed among the users after the period of implementation. The questionnaire was graded using the Likert scale and was analyzed using mean and standard deviation to arrive at the composite mean of each

characteristic. This analysis supplements the technical evaluation performed above.

POPULATION AND SAMPLING

The target population of the study was the Junior High School students of Pasig Christian Academy for the Academic Year 2026-2027. These respondents were selected as the target population since they are the ones who will be using the developed LearnSync AI. The Junior High School students were considered appropriate for participation in the study because of their current status of acquiring independence, including independence in learning or self-regulated learning. Additionally, this level of students was selected as a target population because of the presence of the necessary academic readiness for the use of such an advanced learning module or system and for the purposes of checking and monitoring the learning progress in relation to the use of the module.

The study used proportionate stratified random sampling technique to select 49 students from the 98 students that were considered the target population. The respondents were divided into groups based on grade level and from these groups of equal percentages of students (50%) were randomly selected depending on the percentage of each level as depicted in Table 1. This method ensured proportional allocation of samples within grade levels and minimized selection bias because of the random allocation process applied.

Table 1.

Population and Sample Distribution

Grade Level	Population (N)	Sample (n)
Grade 7	16	8
Grade 8	31	15
Grade 9	32	16
Grade 10	19	10
Total	**98**	**49**

Note. N is the population of each grade level and n the sample drawn from it using EQ (7). Grade 8 (15.5) was rounded down and Grade 10 (9.5) up to keep the total at 49, so the smaller grade level would not be under-represented.

The sample size for each grade level was determined using the proportionate allocation formula:

EQ (7): $n_h = \frac{N_h}{N} \, x \, n$

Where:

- n_h = Sample size for each grade level
- N_h = Population of each grade level
- N = Total Population
- n = Total Sample Size

For the educator respondents, one educator who was in charge of the area of the subject that was determined during the requirements analysis to be

difficult among the participants was included. For the technical evaluation, 10 information technology practitioners were included. All have a bachelor's degree in Information Technology or Computer Science. They were selected through purposive sampling, which selects participants on the basis of a certain qualification rather than randomly, because the study required evaluators who would be able to assess the platform as software. They were given access to both the Student Portal and the Educator Portal so that they could rate all eight quality characteristics. Unlike the students and the educator, they did not use the platform for classroom learning; they assessed it as a completed system. While the respondents came from various grade levels, the study did not measure the study's learning outcomes across Grades 7 to 10; all were analyzed as one study population, with learning progress described through changes in knowledge estimates from the BKT, not differences between grade level groups.

EVALUATION INSTRUMENT

The evaluation instruments included backend learning analytics plus researcher-modified questionnaires addressing learning progress, software quality, and user acceptability.

System-derived knowledge records. The system automatically records, for each student, the BKT knowledge estimate for each learning topic after each practice opportunity, the number of opportunities performed, and the module in which each was generated. These records were generated as part of normal

operations, and no special assessment was conducted. They were extracted at the end of the implementation period and formed the basis for Statement of the Problem No. 4. The correctness of the computed values was separately established using the BKT verification procedure

Quantitative instrument. Students, educator, and information technology practitioners have evaluated the acceptability of the developed software system using a questionnaire formulated based on the ISO/IEC 25010 Software Quality Model consisting of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Students have rated the student-facing features, whereas the educator assessed only the educator module, including the functions of resource upload and management, grade-level assignment, and student learning statistics. Similarly, the practitioners rated the two portals considering all eight characteristics from a technical perspective. The eight-characteristic model of ISO/IEC 25010:2011 was used because it is the model in the college's System Evaluation Tool and has been recently used in Philippine evaluations of similar software (Alan et al., 2026). The respondents scored each item in the questionnaire using a four-point Likert scale, excluding the neutral choice to necessitate an ultimate evaluation: 4 (Strongly Agree or Excellent), in which the functionality consistently performs to expectation; 3 (Agree or Good), which performs satisfactorily but with only a few shortcomings; 2 (Disagree or Fair), which performs partially to expectation with several limitations; and 1 (Strongly Disagree or Poor), which does not perform to expectation.

DATA COLLECTION METHOD

The data was gathered in four distinct phases: conducting a requirement analysis, pre-implementation investigation, implementation, and post-implementation assessment. Each of these stages is collectively integrated in order to incorporate the needs of Pasig Christian Academy as determined through the design, development, implementation, and evaluation of LearnSync AI.

Three groups of participants were involved, each contributing to different research questions. During the requirement analysis phase, before system design and construction, the researchers interviewed the school principal and educator and surveyed the Junior High School students to identify the current independent learning process, challenges, and system's functional and non-functional requirements. Since this research instrument aimed to reveal what students need instead of assessing already existing solutions, the survey was open to all students studying in the 7th to 10th grades; it was not limited to the evaluation sample. Out of 98 students in the accessible population, 62 completed it, which makes 63.27 percent response rate. Another sample of 49 students was selected via proportionate stratified random sampling technique and is presented in Table 1 below; it governed the implementation and evaluation phases.

In the pre-implementation phase the researchers obtained permission from the school administration, explained the objectives, procedures, and duration, and showed participants how to access and use the platform.

The implementation phase lasted for three weeks, the first of which was dedicated to presenting the system to the school principal, organizing and uploading the learning materials via Educator Portal, and setting up and distributing the student accounts. The second one was the actual period of using the product when the 49 selected students were provided with an opportunity to learn with the help of LearnSync AI outside the regular classroom hours; out of them, 46 opened the provided material and generated any records, according to Statement of the Problem No. 4, while the other three did not do that, so they lack any estimations. Finally, the last week was reserved for concluding the matter by distributing the evaluation questionnaires and extracting the necessary learning records from the system.

The subject was determined according to the requirements analysis as one regarded as difficult in every grade level, compatible with the platform's concept-based design capabilities. Science was selected by 15 out of the 62 respondents, or 24.19%, as the subject most challenging to learn without a teacher's assistance, second to Mathematics and the only other subject identified across all four grade levels. The school administration requested the uploading and management of the learning resources by a single educator to avoid disruptions to the teaching schedules of the other subjects; thus, one Science educator was assigned to the implementation period. The choice of Science as the subject of the pilot implementation does not limit the application of the platform to one academic discipline, as the educator can upload resources on multiple subjects.

Students then used the platform's core features, including Adaptive Flashcards, Adaptive Test Mode, the AI Learning Assistant, the Feynman Technique and Free Recall, AI-generated games and learning analytics, choosing freely whether and how to use them, as self-directed study implies. Throughout the period their use of practice opportunities and module level activity was logged, along with the system's BKT estimates for each topic, and the researchers monitored the implementation and provided technical assistance where needed.

In the post-implementation stage, the evaluators completed the ISO/IEC 25010 questionnaire via Google Forms. 57 evaluators participated: the 46 students who experienced the platform, the one educator who took care of the learning materials, and 10 information technology practitioners who assessed the platform from a technical perspective. Their answers were utilized for the statistical analysis of the software quality and the support of independent learning, and the extracted BKT records were analyzed using descriptive statistics to describe learning progress across practice opportunities.

ETHICAL CONSIDERATIONS

This study was conducted under ethical research standards to protect the rights, privacy, and welfare of all participants involved. The permission of Pasig Christian Academy administration was requested and acquired before the commencement of the research. In addition, the participants were informed of the

study's nature, procedure, duration, and purpose; therefore, their voluntary participation was attained. A participant could withdraw from the research anytime he or she wanted.

The study was compliant with the Data Privacy Act of 2012 (Republic Act No. 10173) because it gathered and utilized only the necessary information for the study and to implement the proposed solutions. The information collected was treated as confidential and used solely for research and evaluation. In addition, the reports, presentations, and other documents generated from the study did not include any information that could identify study participants. The information collected was also protected and was not shared with unauthorized third parties. To guarantee the privacy and academic integrity of the students, only authenticated educators could upload learning resources and materials onto the system, and only the student and the authenticating educator could access their learning analytics and personal information.

SUMMARY

This chapter described the methodology of the study which comprised research design, system architecture, algorithm and software engineering methods, data flow and accuracy testing tools, population and sampling procedures, evaluation instrument, data collection procedures, and ethical considerations.

The study utilized a developmental research design involving quantitative methods. LearnSync AI was constructed with the help of the Agile Scrum framework and implements the classical four-parameter Bayesian Knowledge Tracing model to calculate students' knowledge states. During the educator's work, he/she uploads the material available to students and supervises their progress. Adaptive Flashcards and Test Mode update the knowledge tracking states, while AI-assisted strategies, AI-driven games, and individual activities use these states to facilitate self-learning. The portal offers the educator an analytical overview of the class performance based on grades at a grade level.

The study was conducted at Pasig Christian Academy, wherein the Juniors High School students were selected using proportional stratified random sampling and their assigned educator. Three weeks were dedicated to implementing the system, and data was gathered using questionnaires based on the ISO/IEC 25010 Software Quality Model. The correctness of implementation of the BKT was verified against manually calculated estimates. The gathered data was then analyzed using frequency count, percentage, weighted mean and standard deviation with paired-samples t-test with Cohen's d for the change in knowledge estimates, which was evaluated at the .05 level, with ethical principles including compliance with the Data Privacy Act of 2012, was observed throughout.

Using the methodology described in the chapter, the following chapter contains the results of the requirements analysis, development, implementation, and evaluation of LearnSync AI, organized by the Statements of the Problem

followed by a discussion of the results in regards to the objectives and research questions of the study.

CHAPTER III

INTRODUCTION

This chapter presents the results obtained from the conduct of the study on LearnSync AI. The objective of the study was to design, develop, implement, and evaluate an independent learning platform that integrates Generative Artificial Intelligence and Bayesian Knowledge Tracing to provide personalized support, adaptive learning experiences, and continuous monitoring of the learning progress of Junior High School students.

The results have been organized according to the Statements of the Problem formulated in Chapter I. This chapter presents the current independent learning practices and problems of the students, the characteristics incorporated in the platform to deal with them, the verification of the Bayesian Knowledge Tracing engine, the change that has happened in the recorded knowledge estimates, and the level of acceptability of the platform among the targeted end users.

CURRENT INDEPENDENT LEARNING PRACTICES AND CHALLENGES

A researcher-made requirements analysis questionnaire was administered through Google Forms. This consisted of a demographic part, an item to identify what the respondent considered most difficult to understand during unsupervised

study, and twelve close-ended statements that were scored using a four-point Likert scale. Consistent with Chapter II, the scale removed the neutral option to encourage a more definite response. Table 2 below shows the scale and the mean ranges.

Table 2.

Four-Point Likert Scale and Its Interpretation

Scale	Mean Range	Verbal Interpretation
4	3.26 – 4.00	Strongly Agree
3	2.51 – 3.25	Agree
2	1.76 – 2.50	Disagree
1	1.00 – 1.75	Strongly Disagree

Note. The table shows the four-point scale used in the survey and the score range for each interpretation.

Since the requirement analysis was done during the pre-development stage and, therefore, prior to designing the platform, and the objective was to establish requirements rather than to evaluate an existing system, the questionnaire was administered to the entire target population, i.e., Grades 7 to 10, without limiting the respondents to the evaluation sample. Out of the total number of 98 students in the accessible population, 62 students (63.27 percent response rate) participated in the study. The sample of 49 students that was selected through proportionate stratified random sampling and is presented in Table 1 below was used to conduct the study and evaluate the platform and is reported in Statements of the Problem No. 4 and No. 5.

Profile of the Requirements Analysis Respondents

Table 3 below presents the number and percentage of respondents per grade as well as the percentage of the target population per grade.

Table 3.

Survey Respondents and Response Rate by Grade Level

Grade Level	Population (N)	Respondents (f)	Percentage of Respondents (%)	Response Rate (%)
Grade 7	16	9	14.52	56.25
Grade 8	31	24	38.71	77.42
Grade 9	32	17	27.42	53.13
Grade 10	19	12	19.35	63.16
Total	**98**	**62**	**100.00**	**63.27**

Note. n = 62. The table shows how many students from each grade level answered the survey. Percentage of Respondents is the share of the 62 respondents who came from that grade level, and Response Rate is the share of that grade level who responded.

Grade 8 represented the largest percentage of respondents with 38.71 percent and had the highest participation at 77.42 percent of its level, followed by Grade 10 at 63.16 percent, Grade 7 at 56.25 percent, and Grade 9 at 53.13 percent. All four grades participated. 36 of the 98 students did not complete the questionnaire, therefore, the results reflect those who chose to respond, not the entire accessible population.

Subjects Identified as Most Difficult During Independent Study

Respondents were asked to identify the subject they find the most difficult to understand without a teacher's assistance when studying independently at home.

Table 4.

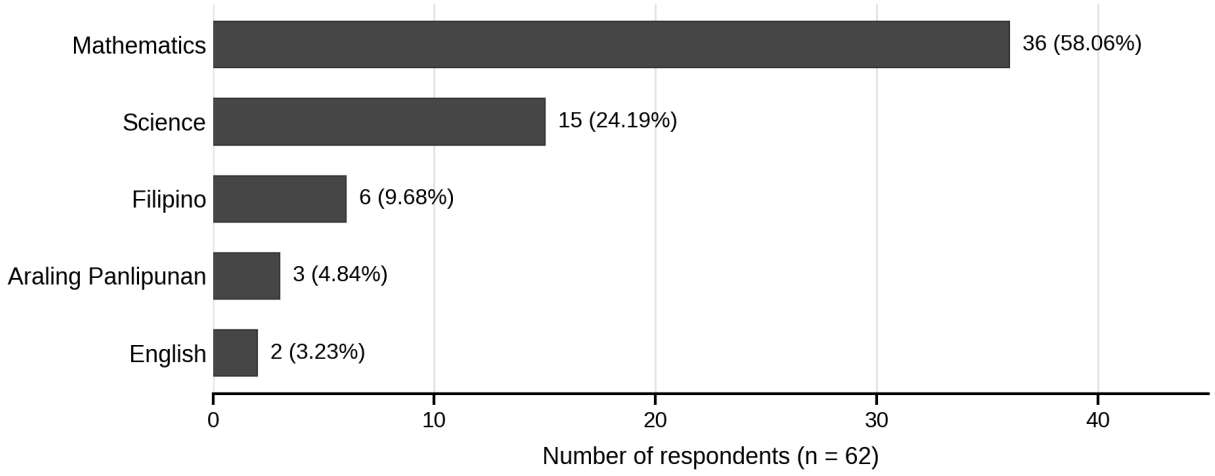
Subjects Students Find Hardest Without a Teacher

Subject Area	Frequency (f)	Percentage (%)
Mathematics	36	58.06
Science	15	24.19
Filipino	6	9.68
Araling Panlipunan	3	4.84
English	2	3.23
Total	**62**	**100.00**

Note. n = 62. The table shows which subject the students find hardest to study alone. Each student chose only one subject.

Figure 5.

Subjects Students Find Hardest Without a Teacher



Note. n = 62. The figure presents the subject results in Table 4 as a bar graph.

Mathematics was identified by 36 out of the 62 or 58.06 percent of the respondents, followed by Science with 24.19 percent, Filipino with 9.68 percent, Araling Panlipunan with 4.84 percent, and English with 3.23 percent. Mathematics and Science were identified by 51 out of the 62 or 82.26 percent of the respondents.

Table 5.

Hardest Subjects by Grade Level

Subject Area	**Grade 7**	**Grade 8**	**Grade 9**	**Grade 10**	**Total**
Mathematics	2	16	11	7	36
Science	5	4	3	3	15
Filipino	0	3	1	2	6
Araling Panlipunan	1	1	1	0	3
English	1	0	1	0	2
Total	**9**	**24**	**17**	**12**	**62**

Note. n = 62. The table shows the same subject choices broken down by grade level. Entries are counts of students.

Mathematics was the most frequently identified subject in Grades 8, 9, and 10, while Science was most frequently identified in Grade 7. Science was also second in Grades 8, 9, and 10, making it the only subject after Mathematics identified across all four grade levels. These results informed the selection of the subject area used as the initial learning content during the implementation phase. Science was designated for this purpose because it is compatible with the concept-based learning capabilities of LearnSync AI established in the scope of the study.

Self-Monitoring and Comprehension Challenges in Independent Study

Table 6 presents the responses of the 62 respondents to the four indicators that describe the difficulties they encounter in monitoring and verifying their own understanding while studying independently.

Table 6.

Difficulty Checking One's Own Understanding

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
3. When reviewing on my own, I find it difficult to identify exactly which specific topics or	18	25	19	0	2.98	0.78	Agree

skills I need to spend more time studying.							
4. When studying independently, I often think I understand a lesson, but I only realize I am wrong when I actually take the test in school.	18	17	23	4	2.79	0.94	Agree
7. I struggle to remember and understand complex lessons when I just read my physical hardbound books without anyone to explain them to me.	15	30	15	2	2.94	0.79	Agree
8. When I try to explain a difficult concept out loud to myself to see if I understand it, I have no way to check if my explanation is actually correct.	16	23	19	4	2.82	0.90	Agree
Composite Mean					**2.88**	**0.62**	**Agree**

Note. n = 62. The table shows how the students answered the four statements about checking their own understanding. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

All four indicators were interpreted as Agree. Difficulty identifying which topics require more study time obtained the highest weighted mean at 2.98, followed by difficulty understanding complex lessons from hardbound books at 2.94 and inability to verify one's own explanation at 2.82. The lowest was the tendency to believe a lesson has been understood until the test proves otherwise, at 2.79, on which 35 respondents (56.45 percent) agreed or strongly agreed and 27 (43.55 percent) disagreed or strongly disagreed. The category obtained a composite mean of 2.88 (SD = 0.62). In frequency terms, 45

respondents (72.58 percent) agreed or strongly agreed that they struggle to understand complex lessons from printed materials without explanation.

Engagement and Motivation Challenges During Independent Study

Table 7 presents the responses to the two indicators that describe the respondents' attention and motivation during independent study.

Table 7.

Difficulty Staying Engaged While Studying Alone

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
5. I lose focus quickly when my independent study relies entirely on reading traditional, plain-text materials.	21	21	14	6	2.92	0.98	Agree
6. I would study longer and more consistently at home if my review materials included game-like features.	30	18	12	2	3.23	0.88	Agree
Composite Mean					**3.07**	**0.75**	**Agree**

Note. n = 62. The table shows how the students answered the two statements about losing focus and staying motivated. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

Both indicators were interpreted as Agree. The statement that game-like features would lengthen and regularize study sessions obtained a weighted mean of 3.23, with 30 respondents strongly agreeing and 48 (77.42 percent)

agreeing or strongly agreeing. Losing focus when studying from plain-text materials obtained 2.92, with 42 respondents (67.74 percent) agreeing or strongly agreeing. The category obtained a composite mean of 3.07 (SD = 0.75).

Limitations of Currently Used Digital Study Tools

Table 8 presents the responses to the four indicators that describe the limitations the respondents encounter in the educational applications and websites they currently use for independent study.

Table 8.

Limits of the Study Apps Students Already Use

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
10. Most educational apps I use just give me a final score but do not automatically adapt to my weaknesses or tell me what specific topics I need to study next.	17	30	11	4	2.97	0.85	Agree
11. The study apps I currently use rely heavily on multiple-choice questions, which makes it easy to guess the answer without actually understanding the lesson.	18	20	20	4	2.84	0.93	Agree

12. I find it frustrating that current educational apps do not allow me to type out or explain my answers in my own words to check if my reasoning is correct.	18	27	11	6	2.92	0.93	Agree
13. I have to use too many different apps or websites because there is no single app that does everything I need for independent study.	25	15	18	4	2.98	0.98	Agree
Composite Mean					**2.93**	**0.65**	**Agree**

Note. n = 62. The table shows how the students answered the four statements about the study apps they already use. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

All four indicators were interpreted as Agree. Having to use several different applications obtained the highest weighted mean at 2.98, followed by applications providing only a final score without adapting to weaknesses at 2.97 and the inability to type or explain answers at 2.92. Heavy reliance on multiple-choice questions, which allows answers to be guessed, obtained 2.84, on which 38 respondents (61.29 percent) agreed or strongly agreed and 24 (38.71 percent) disagreed or strongly disagreed. The category obtained a composite mean of 2.93 (SD = 0.65). Forty-seven respondents (75.81 percent) agreed or strongly agreed that their applications do not adapt to their weaknesses.

Expressed Need for an Integrated and Adaptive Learning Platform

Table 9 presents the responses to the two indicators that describe the respondents' expressed need for a single learning platform that tracks progress and consolidates their independent study requirements.

Table 9.

Need for One Adaptive Study Platform

Indicator	Frequency of Responses				Weighted Mean	Standard Deviation	Verbal Interpretation
	4	3	2	1			
9. I would study more consistently at home if I had an interactive, digital tool that could track my progress and tell me exactly where I need to improve.	31	23	6	2	3.34	0.79	Strongly Agree
14. Having a single, all-in-one application that contains everything I need for independent learning would significantly improve my study habits.	35	18	9	0	3.42	0.74	Strongly Agree
Composite Mean					**3.38**	**0.61**	**Strongly Agree**

Note. n = 62. The table shows how the students answered the two statements about wanting one app that tracks their progress. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

Both indicators were interpreted as Strongly Agree. The statement that a single all-in-one application would significantly improve study habits obtained the highest weighted mean of the entire instrument at 3.42, with 35 respondents

strongly agreeing and none strongly disagreeing. The need for an interactive tool that tracks progress and indicates where to improve obtained 3.34. The category obtained a composite mean of 3.38 (SD = 0.61). Fifty-four respondents (87.10 percent) agreed or strongly agreed with the need for progress tracking, and 53 (85.48 percent) with the need for an all-in-one application.

Summary of the Requirements Analysis Findings

Table 10 presents the composite means of the four categories covered by the requirements analysis questionnaire, and Figure 6 presents the weighted mean of each of the twelve indicators grouped according to these categories.

Table 10.

Summary of Survey Findings by Category

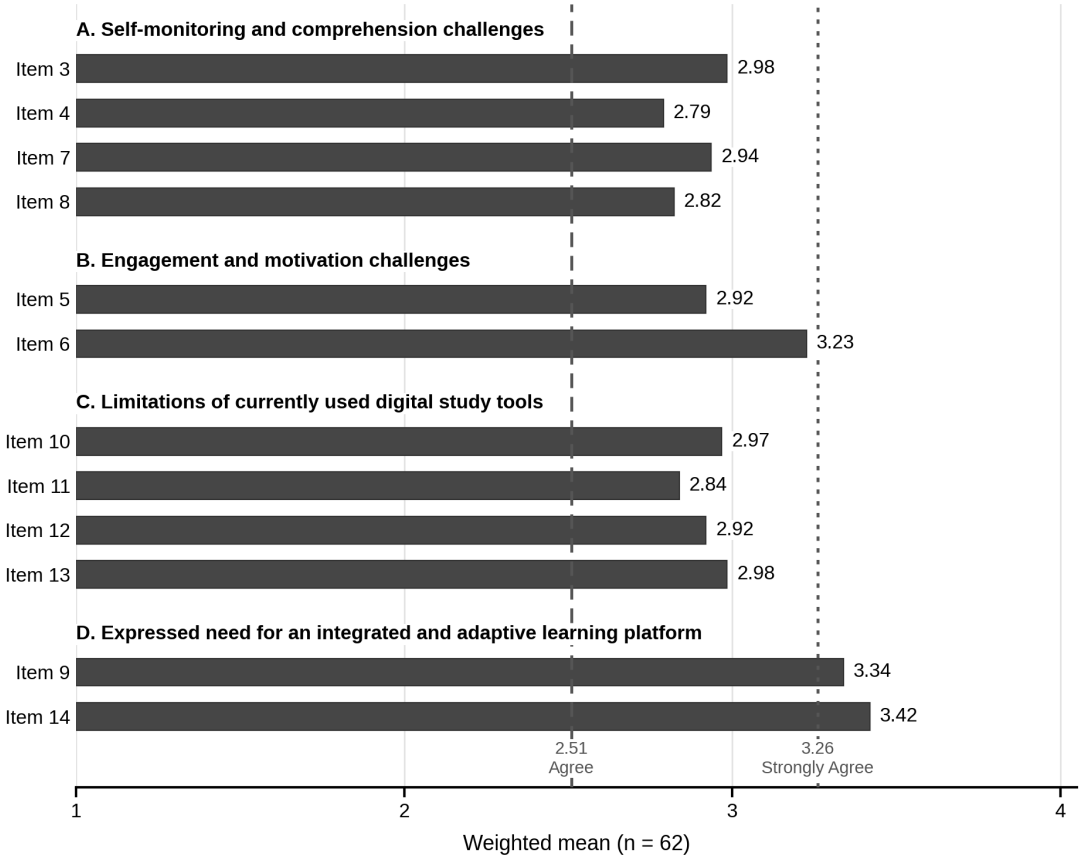
Category	No. of Items	Composite Mean	Standard Deviation	Verbal Interpretation	Rank
A. Self-monitoring and comprehension challenges in independent study	4	2.88	0.62	Agree	4
B. Engagement and motivation challenges during independent study	2	3.07	0.75	Agree	2
C. Limitations of currently used digital study tools	4	2.93	0.65	Agree	3

D. Expressed need for an integrated and adaptive learning platform	2	3.38	0.61	Strongly Agree	1
Overall Composite Mean	**12**	**3.01**	**0.45**	**Agree**	

Note. n = 62. The table summarizes the four groups of statements and ranks them by their composite mean.

Figure 6.

Survey Results by Category



Note. n = 62. The figure shows the weighted mean of each of the twelve statements, grouped by category. The dashed line marks 2.51, the start of Agree, and the dotted line 3.26, the start of Strongly Agree.

The necessity to create an integrated and adaptive platform received the highest score of 3.38 (Strongly Agree), followed by the issues of engagement and motivation (3.07), limitations of the existing tools (2.93), and difficulties in self-monitoring and understanding (2.88), which were all categorized as Agree. The composite mean was 3.01 (SD = 0.45). Twelve out of twelve items had a score above 2.51, while none of the items scored below 2.51 (Disagree/Strongly Disagree).

Discussion

The results suggest that independent learning at Pasig Christian Academy is primarily text-based and unsupervised and that the reported challenges focus on monitoring one's own understanding. The respondents agreed that they cannot accurately determine what they need to study more, check their own explanations, and often only realize their misconceptions during classroom tests. These results could be related to the literature that discusses how learners differ in their abilities to set goals, monitor their progress, and seek guidance during independent learning and that lower levels of self-regulation are linked to poorer outcomes (Briones et al., 2023; Calamlam, 2023; Funa et al., 2023) and the fact that Filipino learners find it challenging to work with self-learning modules where supervision is limited (Abejuela et al., 2023; Agayon et al., 2022).

The identification of Mathematics and Science by 82.26 percent of the respondents suggests that the difficulty lies in conceptual explanation and

procedural reasoning rather than recall. Agreement that plain-text materials cause loss of focus and that game-like review would lengthen study sessions is compatible with literature demonstrating that gamified activities promote attention, motivation, and engagement (Al-Khresheh, 2025; Dahalan et al., 2023). The reported inability to verify one's own explanation is compatible with explanation-based and retrieval-oriented learning research, which finds that explaining material in one's own words promotes comprehension but requires feedback on accuracy to be productive (Corral & Carpenter, 2025; Fiorella, 2023; Serra et al., 2025).

The reported limitations of existing tools, a final score without adaptation, multiple-choice items that permit guessing, and no way to answer in one's own words, parallel the literature contrasting static review systems with adaptive systems that analyze performance to identify weaknesses in real time (Akintola et al., 2025; Lee & Kim, 2024; Onesi-Ozigagun et al., 2024). The concern about guessing is also addressed in knowledge tracing, where probabilistic models estimate knowledge while accounting for guessing and slipping (Bulut et al., 2023).

The highest rated category was the expressed need for an integrated platform, with 87.10 percent agreeing on the feature of progress tracking and 85.48 percent on the concept of an all-in-one application. The need is substantiated by the literature on learning analytics dashboards, which notes that promoting the visibility of progress facilitates awareness, self-efficacy, and engagement if designed with the purpose of regulation and not just

representation (Fleur et al., 2023; Paulsen & Lindsay, 2024; Yang et al., 2024), and the research on the topic, which indicates that integrated adaptive platforms are appropriate for learners who have not yet developed independent study habits (Barrera Castro et al., 2024; Caliston, 2025; Salva et al., 2022).

In addition to the factors mentioned above, it was identified that a platform is needed, which will estimate and display the level of proficiency in topics, evaluate explanations in the user's own words, involve the user in the review process through game-like tasks, provide a constant visual representation of progress, and bring all of the aforementioned elements together in a single interface. These requirements were compiled into the system's requirements as stated in Statement of the Problem No. 2.

FEATURES AND FUNCTIONALITIES INCORPORATED INTO LEARNSYNC AI

This section presents the features incorporated into LearnSync AI in response to the challenges established under Statement of the Problem No. 1. The requirements analysis established the problems students encounter rather than the features to be built. The findings were therefore translated into system requirements, and the method implemented for each requirement was selected from the approaches reviewed in Chapter I rather than proposed by the respondents.

Derived System Requirements

Table 11 presents the system requirements derived from the requirements analysis findings, together with the indicators that support each requirement and the category under which those indicators were reported in Tables 6 to 9.

Table 11.

System Requirements Drawn from the Survey

Req. ID	Derived System Requirement	Supporting Indicators (Weighted Mean)	Category
R1	Estimation and display of topic-level knowledge so that students can identify which concepts require further study	Items 3 (2.98), 4 (2.79), 10 (2.97)	A, C
R2	Assessment that accepts answers expressed in the student's own words rather than selection alone	Items 11 (2.84), 12 (2.92)	C
R3	Evaluation of the correctness of a student's own explanation of a concept	Items 7 (2.94), 8 (2.82)	A
R4	Gamified and interactive review activities calibrated to the learner	Items 5 (2.92), 6 (3.23)	B
R5	Visible and continuous tracking of learning progress with direction on where to improve	Item 9 (3.34)	D

R6	Consolidation of independent study functions within a single platform	Items 13 (2.98), 14 (3.42)	C, D
R7	Educator-managed learning content with grade-level access control and class-level knowledge monitoring	Requirements analysis interviews	Educator

Note. The table lists the seven system requirements drawn from the survey findings, together with the survey items that support each one. R7 came from the interviews with the principal and the educator, so it has no student item.

Seven requirements were derived. R1 to R6 came from the student questionnaire, while R7 came from the interviews conducted with the school principal and the educator during the same pre-development phase. The requirement with the strongest supporting indicators was R6, drawing from Item 14 at 3.42 and Item 13 at 2.98, followed by R5 from Item 9 at 3.34.

Requirements-to-Features Traceability

Table 12 shows the traceability matrix that maps each feature and functionality of LearnSync AI to the requirements that they fulfill. In addition, the matrix was used to confirm during the analysis and design phase that all derived requirements were covered by at least one implemented feature and that there were no features that did not have a requirement.

Table 12.

Requirements-to-Features Traceability Matrix

Feature / Functionality	R1	R2	R3	R4	R5	R6	R7
Adaptive Flashcard Mode	✓					✓	
Adaptive Test Mode	✓	✓				✓	
Free Recall with AI Evaluation		✓	✓			✓	
Feynman Technique Module		✓	✓			✓	
AI Learning Assistant			✓			✓	
PlayMe Gamification Portal				✓		✓	
Student Dashboard and Learning Analytics	✓				✓	✓	
Study Space Hub						✓	
Educator Portal					✓	✓	✓

Note. A check mark indicates that a feature meets that requirement. The table shows that each requirement is met by at least one feature, and that each feature meets at least one requirement.

All seven requirements are addressed, and no feature was developed without a traceable requirement. R1 is addressed by Adaptive Flashcard Mode, Adaptive Test Mode, and the Student Dashboard, the modules through which the BKT engine updates and displays estimates. R2 is addressed by the identification and fill-in-the-blank formats of Adaptive Test Mode and by Free Recall and the Feynman Technique, which accept free-form input. R3 is addressed by Free Recall, the Feynman Technique, and the AI Learning Assistant, which evaluate or respond to student explanations. R4 is addressed by

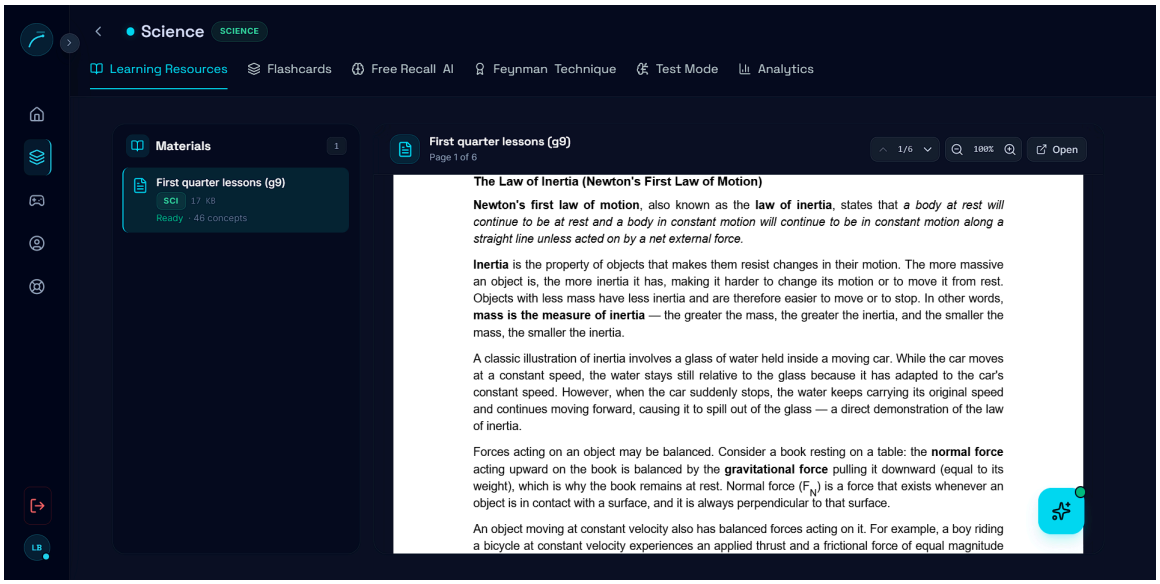
the PlayMe Gamification Portal, R5 by the Student Dashboard and Educator Portal analytics, and R7 by the Educator Portal.

Student Portal Features

The Study Space Hub is the single point of access through which students open materials and select a study mode. Consolidating the modes in one interface addresses R6, which arose from the finding that 64.52 percent of respondents agreed or strongly agreed that they must use several different applications for independent study and 85.48 percent agreed an all-in-one application would improve their habits.

Figure 7.

Study Space Hub

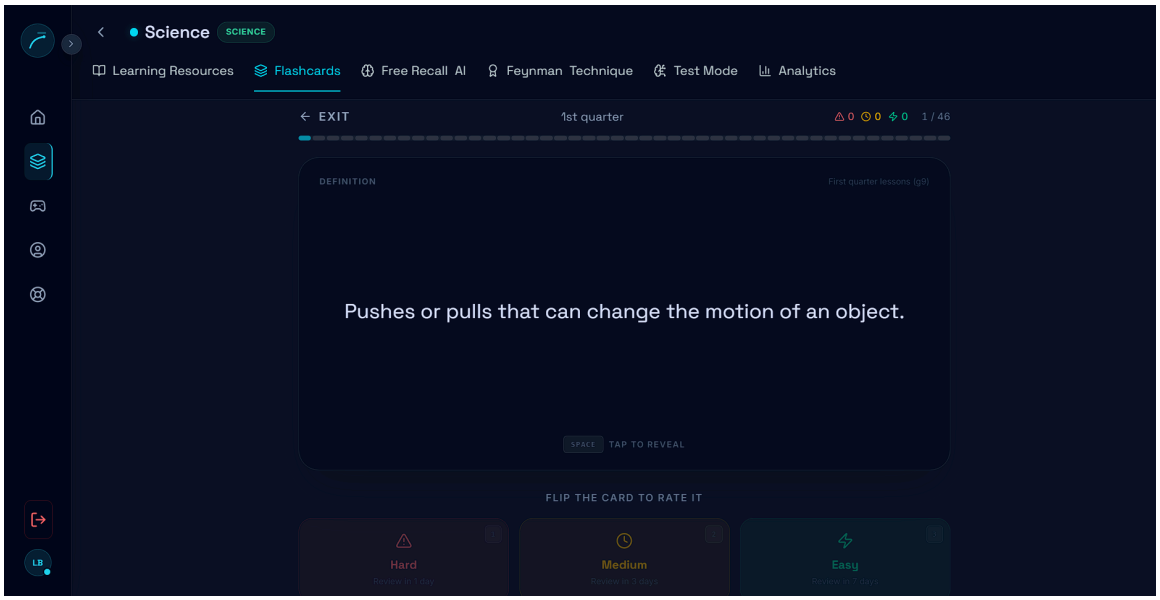


Note. The figure shows the learning resources page of the application wherein the students can view and read the uploaded material of the educator.

Adaptive Flashcard Mode presents concept-level flashcards extracted from the uploaded material and lets the student rate each card as Easy, Medium, or Hard. Every rating is a practice opportunity, and the BKT engine updates the knowledge probability for that concept. Adaptive Test Mode lets the student select from five question formats, grades responses as binary values, and updates the estimate per question; when a new test is requested on the same topic, the stored probability scales the difficulty of the generated items. Together these modes address R1 and, through the open-response formats, R2.

Figure 8.

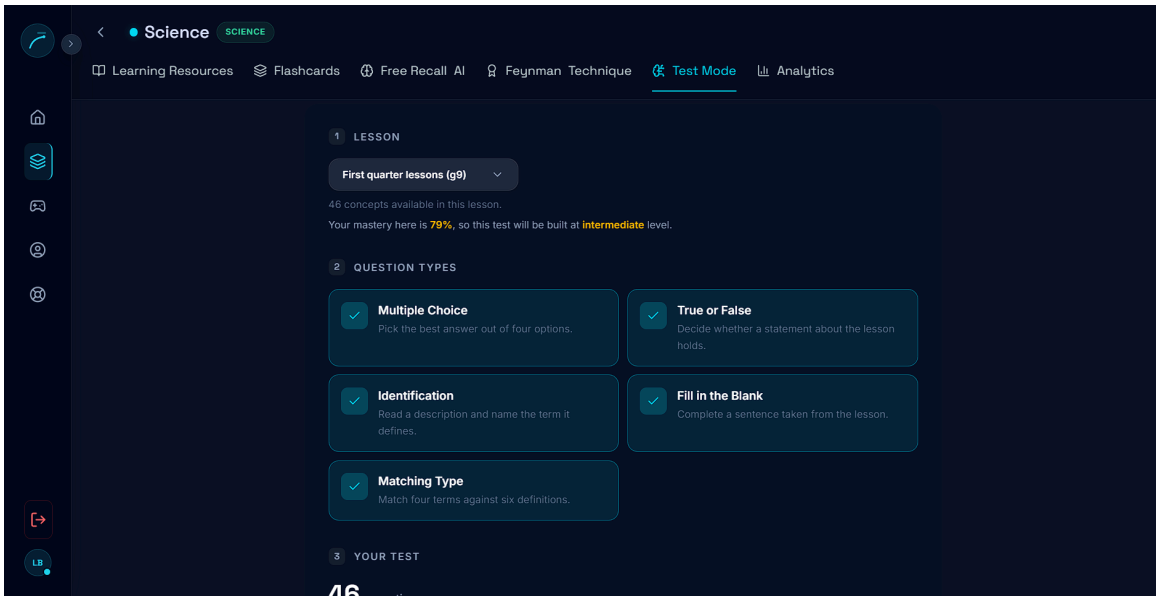
Adaptive Flashcard Mode



Note. The figure shows the flashcard page of the application wherein the students can review the learning materials using flashcards and rate the difficulty of each card.

Figure 9.

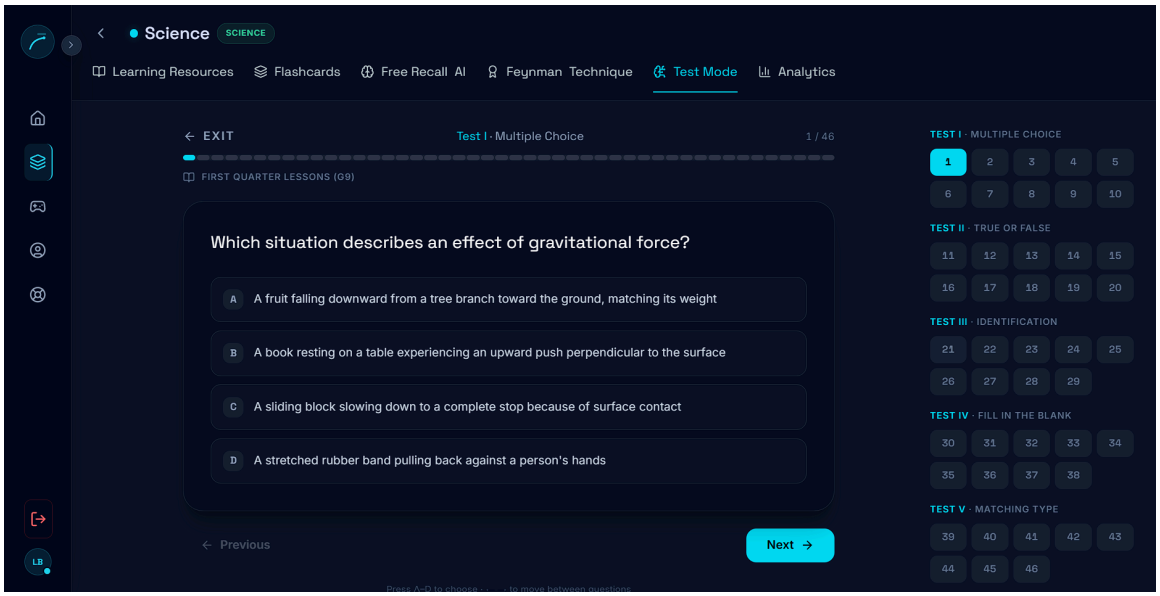
Adaptive Test Mode: Choosing a Format



Note. The figure shows the test mode page of the application wherein the students can select which mode of questions will the generated test contain.

Figure 10.

Adaptive Test Mode: Taking the Test



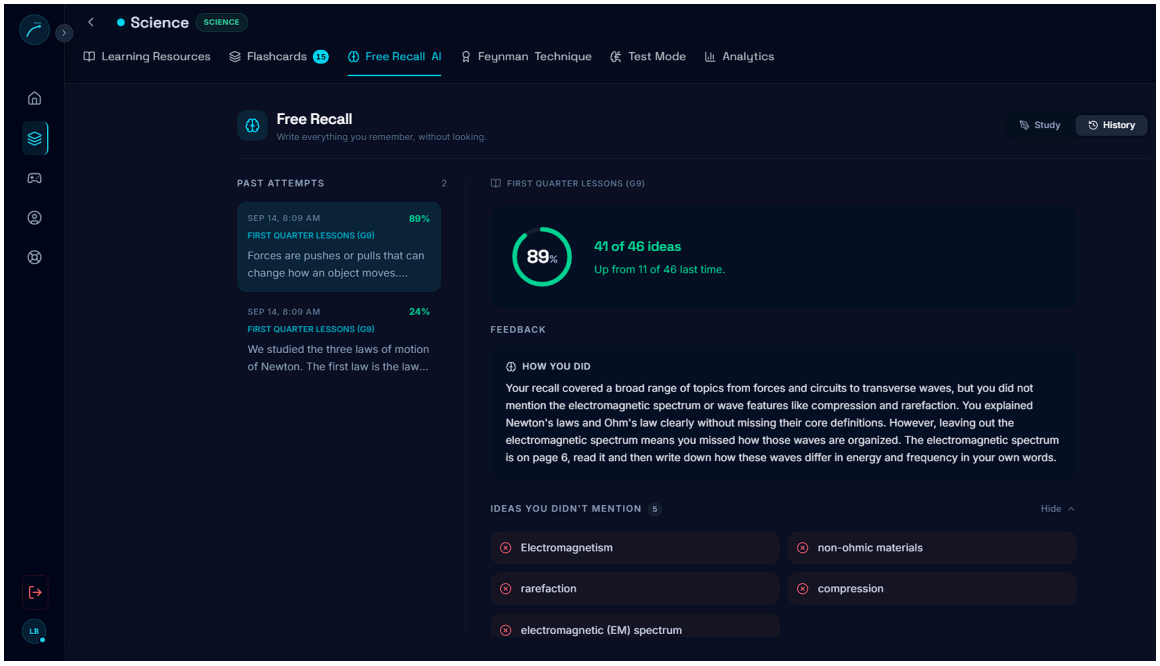
Note. The figure shows the test mode page of the application wherein the students can now answer the generated test that the system produces.

The Free Recall module lets students type everything they remember about a lesson without referring to notes. The response is converted into a vector embedding and compared against the stored lesson vectors using cosine similarity computed in the backend. When the score falls below the 0.70 threshold, the system compares the response against the lesson glossary and identifies the missing keywords. The Feynman Technique module asks students to explain a concept in simple language and evaluates it in a single request

returning one structured result covering three checks: Jargon Check, Analogy Verification, and Gap Finding, producing an accuracy score from 0 to 100, a verdict, and written feedback. These modules address R3, which arose from the finding that 62.90 percent of respondents agreed or strongly agreed that they have no way to check whether their own spoken explanation of a concept is correct, and R2, from the finding that 72.58 percent agreed or strongly agreed that current applications do not allow them to explain answers in their own words.

Figure 11.

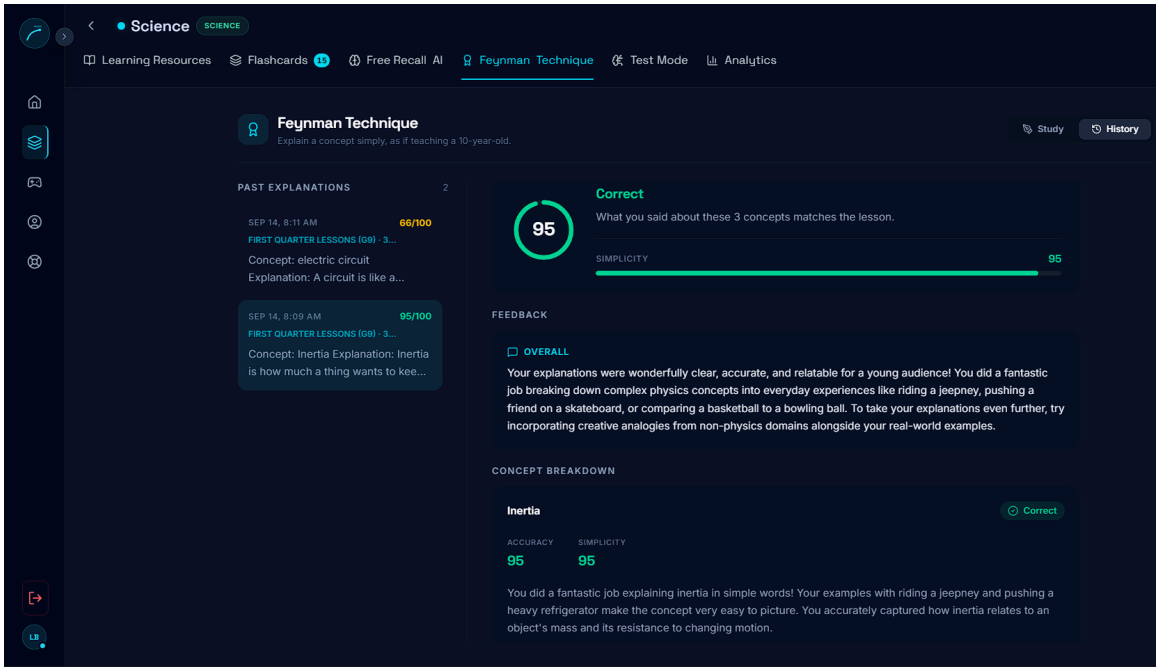
Free Recall Module



Note. The figure shows the Free recall page of the application showing the result of what the student recalled.

Figure 12.

Feynman Technique Module

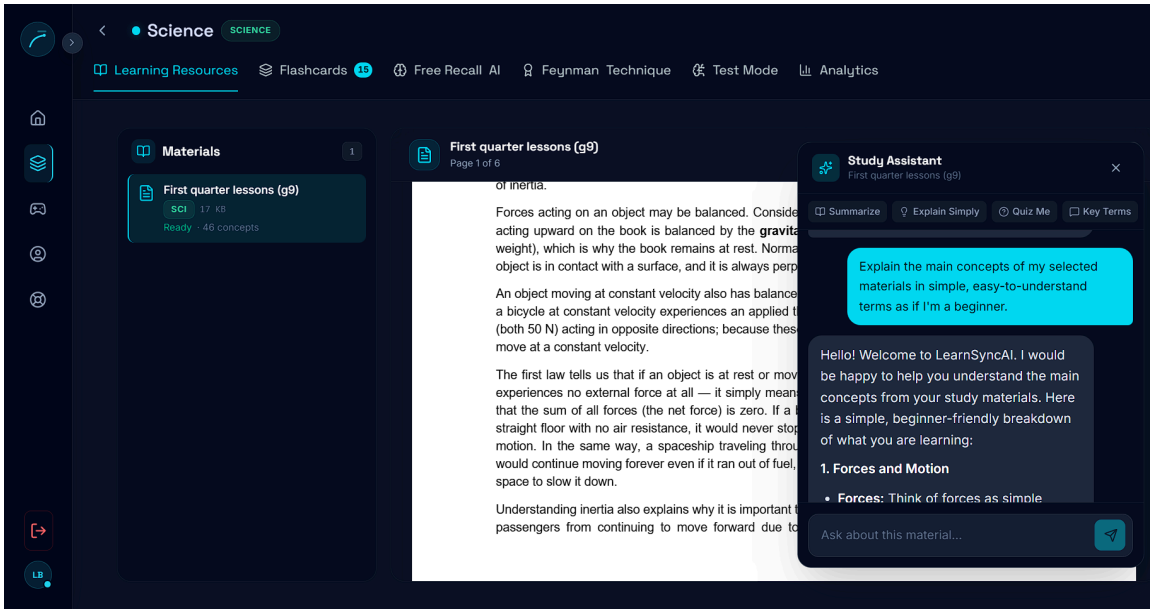


Note. The figure shows the Feynman technique page of the application showing the result of the student's simplified explanation of the selected concept.

The AI Learning Assistant answers student questions with responses grounded in the educator-uploaded resource through vector retrieval, addressing R3 by providing explanations on demand. The PlayMe Gamification Portal provides Crossword, Term Matching, and Timed Trivia, whose vocabulary complexity, clue difficulty, and response timers are constrained by the student's document knowledge index. This addresses R4, which arose from the finding that 77.42 percent of respondents agreed game-like review would lengthen and regularize their study.

Figure 13.

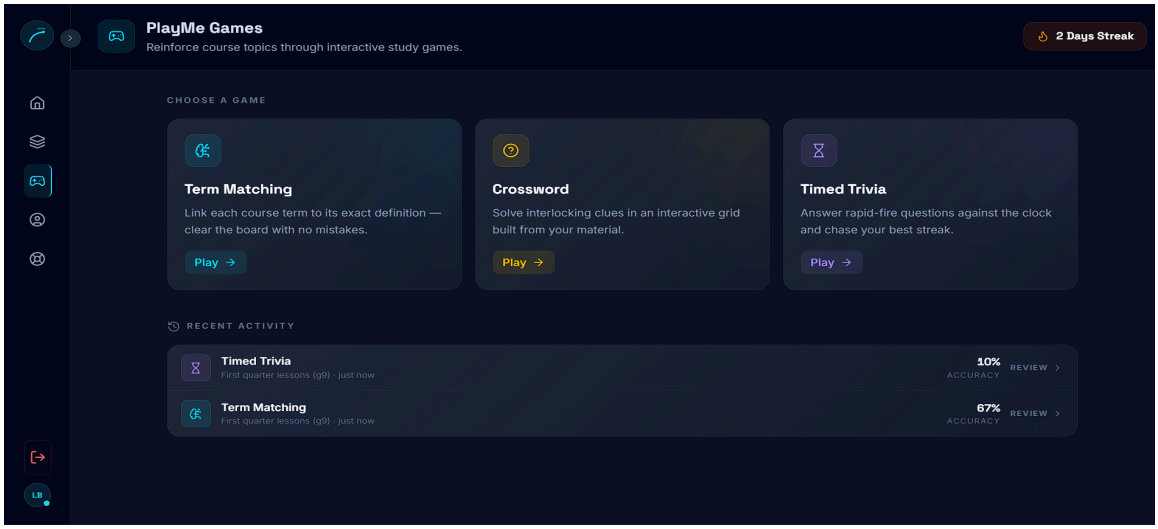
AI Learning Assistant



Note. The figure shows the Learning Resource page of the application showing the assistant's reply to a student's question

Figure 14.

PlayMe Gamification Portal

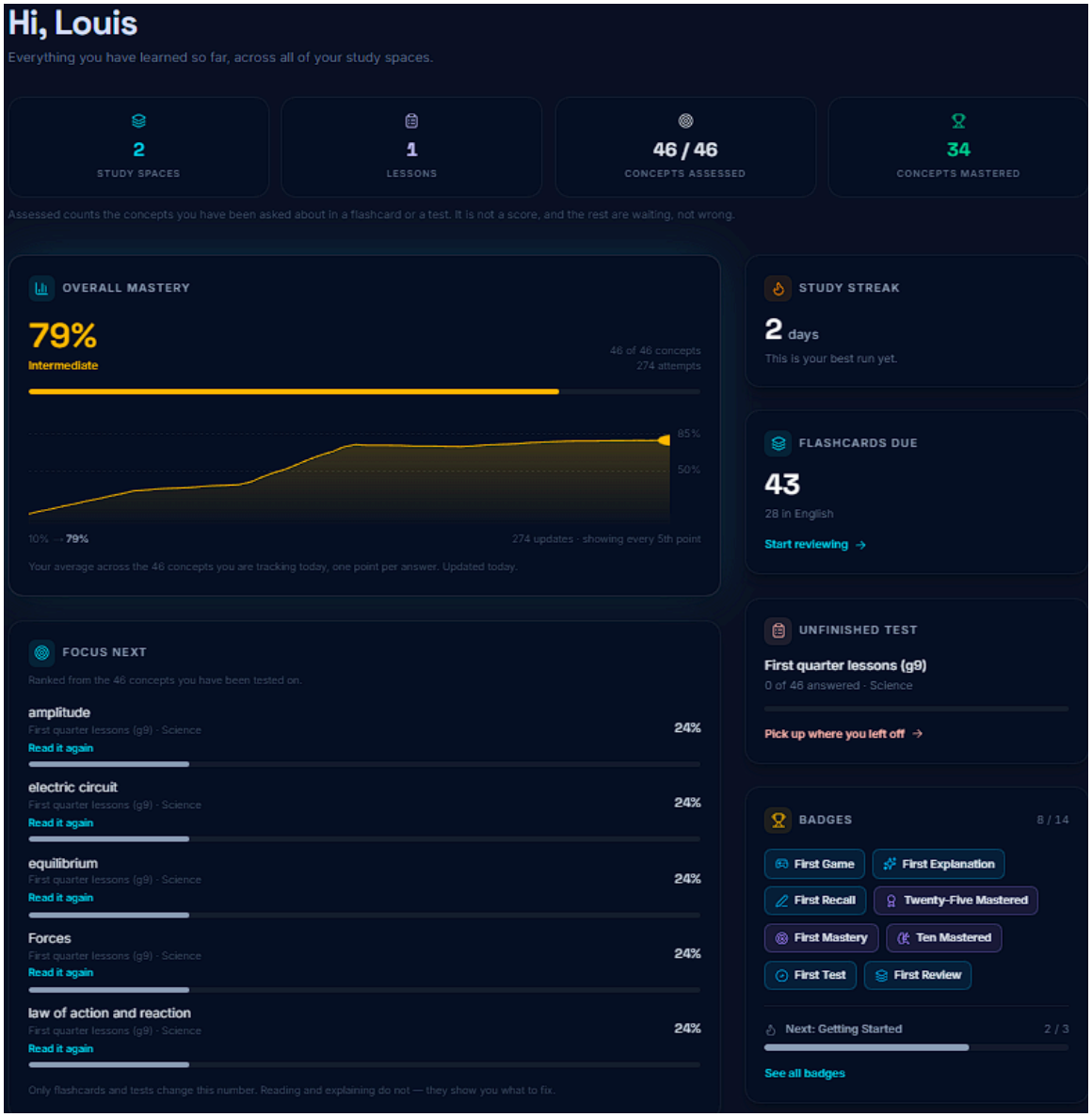


Note. The figure shows the Games page of the application wherein it shows the available games that the application offers.

The Student Dashboard displays the document knowledge index as a progress bar, knowledge curves across practice opportunities, daily streaks, and earned badges. This addresses R5, which arose from the finding that 87.10 percent of respondents agreed an interactive tool tracking progress would make their study more consistent, and supports R1 by making topic-level estimates visible rather than keeping them in the database.

Figure 15.

Student Dashboard



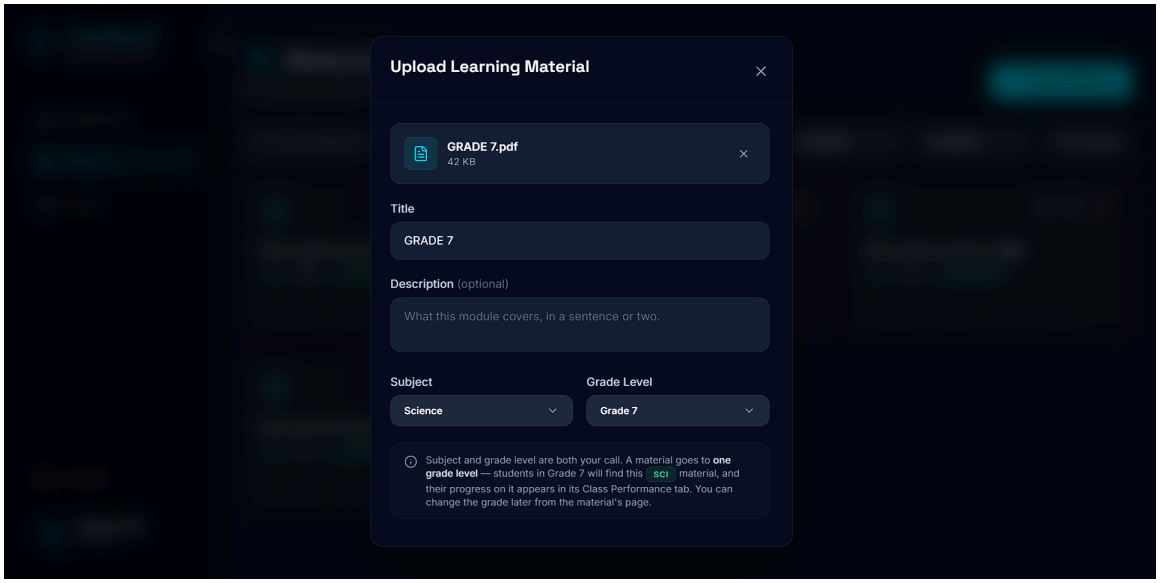
Note. The figure shows the Dashboard page of the student wherein it shows their overall mastery, concepts that they need to focus on, the badges they have earned, and their study streaks

Educator Portal Features

The Educator Portal allows the educator to upload materials as plain-text PDF modules or images of printed pages, to select which grade level may access each material, and to view a real-time analytics overview of student knowledge aggregated at the concept and document level. Image uploads pass through Optical Character Recognition before entering the same parsing pipeline as digital modules, which allows lessons existing only in printed textbooks to be used. These functions address R7 and contribute to R5 by giving the educator a view of class-level progress.

Figure 16.

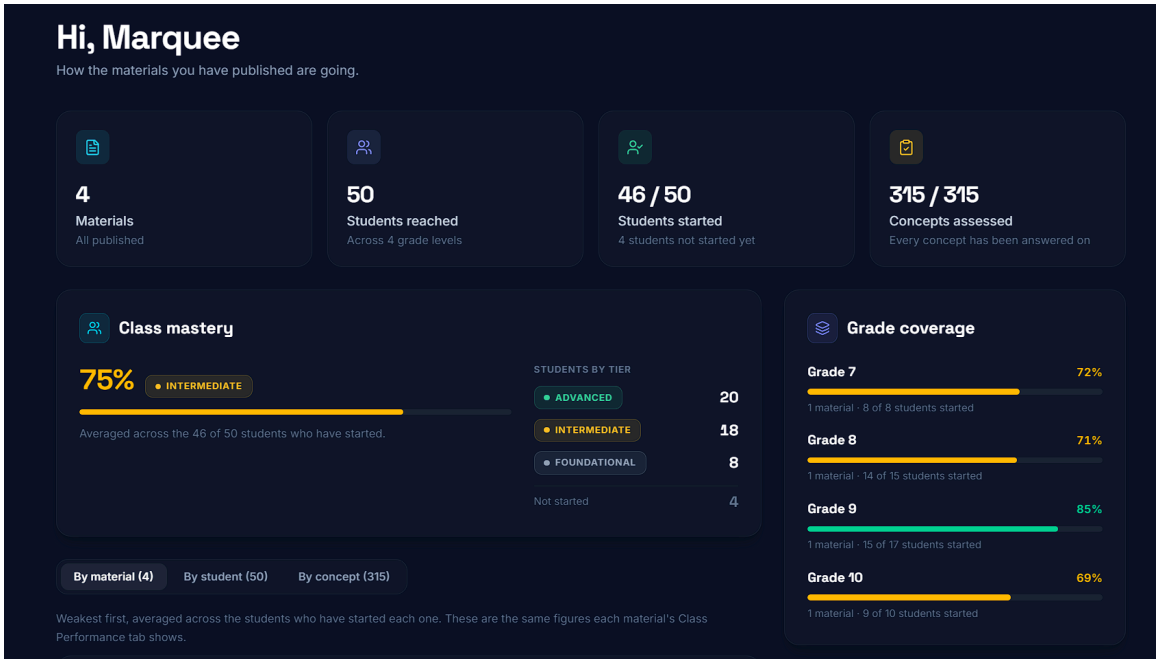
Educator Portal: Uploading Material



Note. The figure shows the upload page wherein the educators can upload their learning materials that the students can use.

Figure 17.

Educator Dashboard



Note. The figure shows the Dashboard page of the educator user wherein it shows the knowledge percentage of their students.

Integration of Bayesian Knowledge Tracing

The classical four-parameter BKT model was integrated as the cognitive modeling core. Each extracted concept is initialized at a baseline prior of 0.10 and updated after every practice opportunity recorded in Adaptive Flashcard Mode and Adaptive Test Mode. Rather than applying one fixed guess probability, the value was calibrated to the response format, since the baseline chance of answering correctly without knowledge differs between a two-option item and an open-ended one.

Table 13.

Guess Probability by Assessment Mode

Assessment Mode	Response Format or Rating	Guess Probability P(G)	Basis for the Value
Adaptive Flashcard Mode	Easy	0.05	High confidence that the correct response reflected genuine knowledge
Adaptive Flashcard Mode	Medium	0.25	Moderate possibility that the response was aided by partial guessing
Adaptive Flashcard Mode	Hard	0.10	Recorded as an incorrect response; produces a steeper decrease in knowledge
Adaptive Test Mode	Multiple Choice (4 options)	0.25	Baseline probability of a correct random response of 1/4
Adaptive Test Mode	True or False	0.50	Baseline probability of a correct random response of 1/2
Adaptive Test Mode	Matching Type (4 terms, 6 definitions)	0.17	Baseline probability of 1/6 applied per matched item
Adaptive Test Mode	Identification	0.05	No provided options; random production of the exact term is negligible
Adaptive Test Mode	Fill in the Blank	0.05	No provided options; free recall of the term without cues

Note. The table shows the guess value used for each answer format.

The guess probability ranges from 0.05 for identification or fill-in-the-blank to 0.50 for true or false. Open-response formats were given the lowest value because they offer no choice, so a correct answer contributes the most to the increase; true or false was given the highest, so a correct binary-choice answer contributes relatively little. This arrangement directly addresses R2, by basing the resulting knowledge gain on whether the student generated the answer or chose it. The aggregated document knowledge index- the arithmetic mean of current probabilities over all the concepts in a document- is presented to the student as a progress indicator and sent to the gamification engine to limit game difficulty.

Integration of Generative Artificial Intelligence

Generative Artificial Intelligence was implemented into seven parts of the platform, namely content ingestion, assessment generation, response evaluation, and gamified content production. Table 14 below displays all the components with their functionalities and the output generated by each part.

Table 14.

Generative AI Components and Their Functions

Generative AI Component	**Function**	**Output**
Document parsing and concept extraction	Extracts key terms, concepts, and definitions from uploaded materials and generates vector embeddings	Concept nodes and glossary used by all study modes

Optical Character Recognition	Converts images of printed textbook pages into machine-readable text	Text that enters the same parsing pipeline as digital modules
Adaptive test item generation	Generates practice items in five formats, scaled by the running BKT knowledge probability	Advanced tier at P(L) of 0.85 and above, intermediate at 0.50 to 0.85, foundational below 0.50
Free Recall evaluation	Compares the student's typed recall against the stored lesson vectors and glossary	Similarity score and a list of missing keywords, threshold 0.70
Feynman Technique evaluation	Applies one structured evaluation covering Jargon Check, Analogy Verification, and Gap Finding	Accuracy score of 0 to 100 and a verdict of Correct, Needs Improvement, or Incorrect
AI Learning Assistant	Answers student questions through vector retrieval from the uploaded resource	Responses grounded in the educator's material
Procedural game generation	Generates crosswords, term matches, and trivia constrained by the document knowledge index	Game content calibrated to the student's knowledge level

Note. The table shows the seven components of the system based on Generative AI and the products that each of them generates. All components are based only on the materials that the educator uploaded.

The components function as a connected pipeline and not as standalone utilities. The parsing component produces the concept nodes and the glossary that all other components use as a foundation: The test generator selects the items from the concepts, the Free Recall evaluator compares the response to the stored vectors and glossary, the Feynman evaluator compares the jargon to the same glossary, the AI Learning Assistant pulls from the same embeddings, and the game generator picks the words from the same parsed content. In other

words, everything the student receives as an input is based on the materials that the educator uploaded rather than the model's knowledge.

The Generative AI components were verified on 37 outputs using the four criteria described in Chapter II. Table 15 below presents the results for each component.

Table 15.

Generative AI Verification Results by Component

Component	Outputs Checked	Relevant	Correct	Matches Tier	Consistent
Document parsing and concept extraction	11	10 of 10	10 of 10		1 of 1
Optical Character Recognition	2		1 of 1		1 of 1
Adaptive test item generation	6	5 of 5	5 of 5	5 of 5	1 of 1
Free Recall evaluation	4	3 of 3	3 of 3		1 of 1
Feynman Technique evaluation	4	3 of 3	3 of 3		1 of 1
AI Learning Assistant	6	5 of 5	4 of 5		1 of 1
Procedural game generation	4	3 of 3	3 of 3	3 of 3	1 of 1
Total	37	29 of 29	29 of 30	8 of 8	7 of 7
Pass rate (%)		**100.00**	**96.67**	**100.00**	**100.00**

Note. Each cell shows the number of outputs that passed out of the outputs rated on that criterion. A dash means the criterion does not apply to that component. Consistency compares one repeated run with the first run.

All relevance, difficulty, and consistency tests passed, and 29 out of 30 outputs (96.67 percent) were factually correct in relation to the uploaded material. The only error was in the AI Learning Assistant responding to a query about transform boundaries by stating that the information is not present in the material when, in fact, the topic is covered in the module. The assistant did not fabricate any responses to the given questions but failed to find content that was in the material.

Discussion

The features were determined by the requirements established in the preceding section rather than selected independently of the identified needs. The traceability matrix in Table 12 shows complete bidirectional coverage, with every requirement addressed by at least one feature and every feature tracing to at least one requirement. This correspondence is consistent with the developmental research design described in Chapter II, in which educational needs are translated systematically into functional features.

Selecting Bayesian Knowledge Tracing as the cognitive core responds to the finding that students cannot identify which topics require further study. The model estimates knowledge per concept while accounting for guessing and slipping, and remains interpretable enough to display to a learner as a mastery estimate rather than a score, properties consistent with the literature describing BKT as a widely used and interpretable approach to learner modeling

(Abdelrahman et al., 2023; Bulut et al., 2023). Varying the guess parameter by response format follows evidence that allowing guess and slip to vary between responses produces more accurate learner models than fixed per-concept values (Šarić-Grgić et al., 2024).

Using Generative AI for adaptive item generation responds to the finding that existing applications report a score without adapting to weaknesses. Conditioning item difficulty on the running knowledge estimate corresponds with literature on adaptive systems that analyze performance to recommend targeted activities (Akintola et al., 2025) and with reported implementations of AI-driven assessment generation (Al-Gburi, 2025; Bahroun et al., 2023). Grounding the AI Learning Assistant in educator-uploaded resources rather than general model knowledge corresponds with approaches that constrain generative responses to an authoritative source to maintain factual alignment (Sajja et al., 2023).

The Free Recall and Feynman Technique modules operationalize two strategies the literature identifies as effective but that students reported being unable to verify alone. Retrieval-based study is reported to be more effective than passive review for retention and metacognitive monitoring (Corral & Carpenter, 2025; Serra et al., 2025), and explanation-based tasks are reported to support comprehension and knowledge generalization when the learner receives feedback on the accuracy of the explanation (Fiorella, 2023). The automated evaluation supplies the feedback component that the respondents identified as missing.

The PlayMe Portal responds to the finding on student concentration loss with plain texts. Adjusting the challenge level of the game based on the current knowledge index is consistent with the reports that gamification activities aid in sustaining attention, motivation, and engagement (Al-Khresheh, 2025; Dahalan et al., 2023; Pellas, 2024) such as playing crosswords as a part of a rigorous review process (Arnold et al., 2024). The Student Dashboard was designed to address the progress-tracking requirement (R5) by providing a visual representation of progress, which has been shown to promote awareness and engagement when focused on promoting regulation rather than simply providing information (Fleur et al., 2023; Paulsen & Lindsay, 2024; Yang et al., 2024). These features comprise the developed system whose accuracy in recording the progress and overall acceptability was reported in Statements of the Problem No. 3, No. 4, and No. 5.

VERIFICATION OF THE BAYESIAN KNOWLEDGE TRACING ENGINE

This section contains the verification performed over the BKT engine. Following the deterministic algorithm verification described in Chapter II, the researchers verified that the knowledge estimates produced by the deployed system are mathematically equal to those that would have been generated had the same equations been applied outside the application to the same responses. The verification was in two parts: 46 constructed test cases run inside the live

application (Part A), and the entire population of updates recorded during the implementation period (Part B).

Verification Procedure

The engine has one knowledge probability for each student and concept which is updated after every practice opportunity using EQ (1) or EQ (2) with EQ (3). The parameters used were $P(L_0) = 0.10$, $P(T) = 0.20$, and $P(S) = 0.15$, with context-sensitive guess probabilities in Table 13, where Easy and Medium flashcard ratings are entered as correct observations and Hard as incorrect. One test case is one update, and a case passed if the absolute difference between system and manual values were within the ±0.01 tolerance established in Chapter II. Correctness was quantified using EQ (6).

Expected values were computed in a companion spreadsheet, in which every value is a visible formula. In Part A, only two categories of information were taken from the system as inputs: the graded observation sequence and the parameter table. Every sequence was recomputed forward from 0.10, so that the expected prior at any step was the expected posterior of the preceding step. The priors stored by the application were never used as inputs in Part A but were treated as additional values to be verified, so that an engine with a defect could not produce apparent agreement by supplying its own intermediate values. Part B applied the two checks described in its own section. System values were read

from the audit table, which records a single row for each update, and from the table holding the current estimate and accumulated opportunity count.

The eight guess values and the binary observation produce thirteen distinct update branches. Table 16 presents the number of updates falling under each branch in the two parts.

Table 16.

Coverage of the Thirteen Update Types

Update Branch	P(G)	Part A	Part B
Multiple choice · correct	0.25	23	1,924
Multiple choice · incorrect	0.25	23	1,144
True or false · correct	0.50	12	2,153
True or false · incorrect	0.50	34	866
Identification · correct	0.05	34	661
Identification · incorrect	0.05	12	404
Fill in the blank · correct	0.05	23	149
Fill in the blank · incorrect	0.05	23	114
Matching type · correct	0.17	22	368
Matching type · incorrect	0.17	22	288
Flashcard Easy · correct	0.05	16	1,494
Flashcard Medium · correct	0.25	15	1,328
Flashcard Hard · incorrect	0.10	15	1,174
Total		**274**	**12,067**

Note. The table shows how many updates of each type were checked in each part. Part A covered every branch by design; Part B shows how often each occurred in ordinary use.

Part A: Designed Test Cases Executed in the Live Application

Part A used a researchers' test account on a material containing 46 extracted concepts. Five adaptive tests were taken, each restricted to one question format, so every concept was answered once in each of the five formats; matching type covers four concepts per block and therefore reached 44 of the 46. Responses were chosen by design according to concept position, producing consecutive correct, consecutive incorrect, alternating, and mixed sequences, after which a flashcard deck from the same material was rated once per card. The application graded all 228 responses exactly as intended, producing 274 updates.

Table 17.

Step-by-Step Check of One Test Case

Step	Format or Rating	Outcome	P(G)	Prior	P(L | obs) EQ (1) / EQ (2)	Expected Posterior EQ (3)	System Value	Difference
1	Multiple choice	Correct	0.25	0.1000	0.2742	0.419355	0.419355	0.0000
2	True or false	Incorrect	0.50	0.4194	0.1781	0.342466	0.342466	0.0000
3	Identification	Correct	0.05	0.3425	0.8985	0.918816	0.918816	0.0000
4	Fill in the blank	Incorrect	0.05	0.9188	0.6412	0.712954	0.712954	0.0000

5	Matching type	Correct	0.17	0.7130	0.9255	0.940382	0.940382	0.0000
6	Flashcard Medium	Correct	0.25	0.9404	0.9817	0.985356	0.985356	0.0000

Note. The prior at step 1 is $P(L_0) = 0.10$. At each later step the prior is the previous step's expected posterior, computed independently of the value stored by the application.

The current estimate stored for this concept was 0.985356 after six practice opportunities, equal to the last expected posterior.

Table 18.

Results of the Forty-Six Designed Test Cases

Designed Sequence	Rating	Concepts	Steps	Updates	Expected Final	System Final	Difference	Result
All correct	Easy	11	6	66	1.0000	1.0000	0.0000	PASS
All correct	Easy	1	5	5	0.9999	0.9999	0.0000	PASS
All incorrect	Hard	11	6	66	0.2406	0.2406	0.0000	PASS
All incorrect	Hard	1	5	5	0.2397	0.2397	0.0000	PASS
Alternating	Medium	11	6	66	0.9854	0.9854	0.0000	PASS
Mixed	Easy	4	6	24	0.9988	0.9988	0.0000	PASS
Mixed	Medium	4	6	24	0.9938	0.9938	0.0000	PASS
Mixed	Hard	3	6	18	0.8900	0.8900	0.0000	PASS
Total		**46**		**274**			**0.0000**	**PASS**

Note. Concepts in the same row received identical designed sequences, so they share the same expected final estimate. Per-concept detail is in the companion workbook.

All 46 designed cases passed. Consecutive correct sequences approached 1.0000, consecutive incorrect sequences settled at 0.2406, and

alternating and mixed sequences resulted in intermediate values based on the guess probability at each step. All 274 updates and all 46 stored final estimates satisfied the ±0.01 tolerance, and all comparisons differed from one another by less than 1×10^{-9}. In each case the prior recorded by the application matched the independently chained prior, and the guess probability applied matched the format-specific or rating-specific value in Table 13.

Part B: Complete Population of Recorded Updates

Part B expanded the comparison from the designed cases to all updates produced by normal student use. All 12,067 updates recorded for the 46 participating students (3,494 student-concept sequences) were recalculated, and two checks were made on every update: first, the posterior was recalculated from the stored prior, recorded observation, and applicable parameters, and compared to the stored posterior; second, the stored prior was checked against the stored posterior of the previous update for the same student and concept, or against 0.10 at the first update. The two checks together are equivalent to recomputing every sequence forward from 0.10 as in Part A, because any stored prior that did not match the preceding result would have failed the second check. All 12,067 updates passed both checks and applied the correct guess probability for the recorded format or rating.

Implementation Correctness Rate

Table 19 presents the Implementation Correctness Rate computed for each verification part using EQ (6), together with the agreement between the current knowledge estimate stored for each sequence and the last expected posterior computed for that sequence.

Table 19.

Correctness Rate by Verification Part

Verification Part	Sequences	Test Cases	Passed	Largest Difference	ICR (%)	Final Estimates in Agreement
A. Designed test cases executed in the live application	46	274	274	0.0000	100.00	46 of 46
B. Complete population of recorded updates	3,494	12,067	12,067	0.0000	100.00	3,494 of 3,494

Note. One test case is one knowledge-estimate update. A case passed when the system and manual values differed by no more than ±0.01, following EQ (6).

The Implementation Correctness Rate was 100.00 percent, in both cases (All 274 designed test cases of part A, corresponding to the procedure described in Chapter II, passed, and extending the comparison to the whole population of 12,067 updates in part B yielded the same result). The largest absolute deviation observed anywhere was 0.0000, which is agreement to the limit of double-precision arithmetic, rather than just being within tolerance. The stored estimate for every sequence was equal to the last expected posterior with a matching opportunity count, so we have established that the values shown to

students and used to calibrate item generation and game difficulty are in fact, the values produced by the verified update sequence.

Discussion

The verification confirms that the BKT engine correctly implements the standard four-parameter equations. As with Chapter II, this is not a statement on the accuracy of predictions, but on the correctness of the implementation. Perfect correctness with minuscule deviation is the expected rather than surprising result of such a verification because the engine and the workbook evaluate the same closed-form equations in double-precision arithmetic and a significant difference would reflect a defect.

The value of the procedure is defined by the four specific defects it was designed to catch: a guess probability that does not match the question format or flashcard rating; a missing transition phase after the evidence phase; an incorrect prior read at the beginning of an update; or a displayed estimate that differs from the estimates that produced it. Each would have created a deviation far above the ±0.01 tolerance, and none were observed.

The results further confirm that the guess parameter configuration in Table 13 is operating as specified since the probability applied corresponds to the recorded format or rating across all 12,067 updates. This is consistent with the finding that allowing guess and slip to vary between responses produces more accurate learner models than using fixed per-concept values (Šarić-Grgić et al.,

2024) and confirms that the engine is indeed performing as described in the four-parameter formulation in the literature (Bulut et al., 2023).

Three conditions apply to interpretation. What entered the engine is determined by the application's grading component rather than by the researchers, and there was no discrepancy between designed response and recorded observation in Part A. The test account used in Part A is not part of the study cohort and is not included in the records cited under Statement of the Problem No. 4. The process is reproducible, as Part B covers every recorded update (as opposed to a sample) and individual examination of every formula in the companion workbook is possible. Thus, having verified that the recorded estimates are calculated correctly, the values recorded during the implementation period may be taken as a faithful record of what the engine measured.

CHANGE IN THE RECORDED KNOWLEDGE ESTIMATES

This section shows the change seen in the BKT knowledge estimates captured by LearnSync AI throughout the one-week usage period carried out between the 7th and 13th of September 2026, which is the second week of the three week implementation described in Chapter II. The figures shown are the platform's own captured output, extracted by the educators' mastery evidence export and verified for arithmetic correctness under Statement of the Problem No. 3.

What the Knowledge Estimate Measures

The engine stores an estimate for each student and concept, representing the student's probability of having learned that concept. It only updates the estimate when a practice opportunity is recorded (when a student responds to an adaptive test question, or rate a due flashcard). Two values were read from each student-concept pair: the estimate after the first practice opportunity, as recorded in the first update, and the current estimate as it stands now. A student's figure at either point is the mean of the estimates for the concepts they practiced, and the change is the paired difference between the two means across the same concepts.

The estimate after the first opportunity is not a baseline and is not a pre-test measurement. The outcome of the first response is already contained within it, so a first correct multiple-choice answer produces 41.9 percent while a first incorrect answer produces 21.7 percent. The initialization constant of 0.10 is identical for every student, carries no information about any individual, and is never reported as a measured value. Concepts a student was never asked about are excluded from both means rather than entered at the constant, so the denominator throughout is what each student practiced. This quantity is therefore distinct from the material-level coverage measure in EQ (4), which enters unpracticed concepts at the constant and is not used here.

Participation and Volume of Practice

Table 20 presents the participation and the volume of practice recorded for each grade level during the implementation period.

Table 20.

Participation and Volume of Practice by Grade Level

Grade	Concepts	Enrolled	Started	Non-Users	Adaptive Tests	Test Questions	Card Ratings	Opportunities
Grade 7	102	8	8	0	18	1,836	799	2,635
Grade 8	97	15	14	1	33	3,201	1,449	4,650
Grade 9	46	16	15	1	34	1,564	1,042	2,606
Grade 10	70	10	9	1	21	1,470	706	2,176
All	**315**	**49**	**46**	**3**	**106**	**8,071**	**3,996**	**12,067**

Note. A practice opportunity is one recorded knowledge-estimate update. Non-users are students who never opened the material and are excluded from every mean in this section.

Forty-six of the 49 enrolled students used the platform, generating 12,067 practice opportunities across 3,494 student-concept pairs. Of these, 8,071 came from adaptive-test questions across 106 test attempts and 3,996 from flashcard ratings. Every student who started the material was assessed on all of its concepts, so the number of pairs per grade is the product of started students and concepts. Three enrolled students never used the platform and carry no estimate.

Change Across Successive Practice Opportunities

Table 21 presents the mean knowledge estimate after each successive practice opportunity on a concept, computed over every student-concept pair that reached that opportunity, together with the distribution of those pairs across the platform's three mastery tiers.

Table 21.

Knowledge Estimate After Each Practice Opportunity

Opportunity Index	Pairs Observed	Mean Estimate	Increase From Previous	Foundational	Intermediate	Advanced
1	3,494	34.4%		88.4%	11.6%	0.0%
2	3,064	58.1%	+23.7 pts	39.9%	32.2%	27.9%
3	2,508	74.7%	+16.6 pts	22.7%	22.2%	55.1%
4	1,760	83.9%	+9.2 pts	12.9%	15.8%	71.3%
5	888	88.5%	+4.6 pts	9.6%	9.2%	81.2%
6	289	89.5%	+1.0 pts	9.3%	8.0%	82.7%

Note. The opportunity index counts practice opportunities on a concept, so index 1 is the first graded one. Indices beyond six are omitted as fewer than thirty pairs reach them.

The mean estimate rose at every successive opportunity, from 34.4 percent to 89.5 percent. The increase was steepest between the first and second

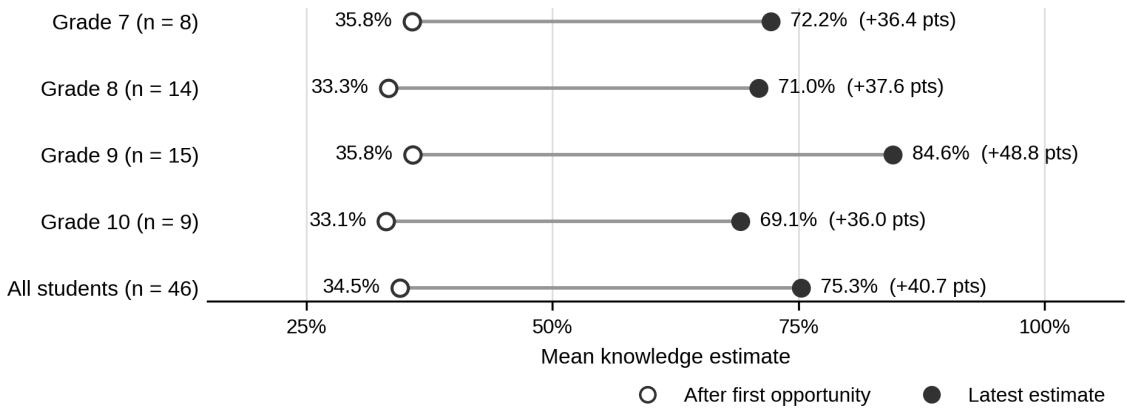
opportunity at 23.7 percentage points and diminished as estimates approached the upper limit of the scale, reaching 1.0 point between the fifth and sixth. The tier distribution shifted correspondingly: no pair could reach the advanced tier after a single opportunity, since the largest posterior obtainable in one step from 0.10 is 72.3 percent, whereas 71.3 percent of pairs were advanced by the fourth. These means are computed over a diminishing subset, since only 50.4 percent of pairs reached a fourth opportunity and 8.3 percent reached a sixth; the values at higher indices therefore describe the concepts students continued to practice rather than the full set.

Change per Student Over the Implementation Period

Figure 18 presents the mean knowledge estimate per student, averaged within each grade level, after the first practice opportunity and at the end of the implementation period

Figure 18.

Knowledge Estimates at the Start and End, by Grade Level



Note. n = 46 started students. Each value is the mean, within a grade level, of the students' own mean estimates over the concepts they practiced; the final row covers all started students. Non-users are excluded. The grade levels are sampling strata; no test was conducted between them.

Table 22 presents the paired change per student, tested using a paired-samples t-test on the 46 started students and on each grade level separately. Statistical significance was evaluated at the .05 level.

Table 22.

Change in Knowledge Estimate per Student, by Grade Level

Group	n	After First Opportunity	Latest Estimate	Mean Change (pts)	Standard Deviation	Median Change	Range	t	p	dz
Grade 7	8	35.8%	72.2%	+36.4	23.1	+38.6	+1.6 to +68.6	4.45	.003	1.57
Grade 8	14	33.3%	71.0%	+37.6	17.6	+45.9	+0.3 to +58.8	8.00	< .001	2.14
Grade 9	15	35.8%	84.6%	+48.8	14.8	+53.3	+7.5 to +61.2	12.80	< .001	3.30

Grade 10	9	33.1%	69.1%	+36.0	21.4	+38.0	+0.4 to +58.5	5.04	.001	1.68
All started	**46**	**34.5%**	**75.3%**	**+40.7**	**18.8**	**+46.5**	**+0.3 to +68.6**	**14.68**	**< .001**	**2.16**

Note. Change is the latest estimate minus the estimate after the first opportunity, in percentage points, paired within students. The effect size is Cohen's d for paired samples. Each test is a within-student paired comparison; no test was conducted between grade levels.

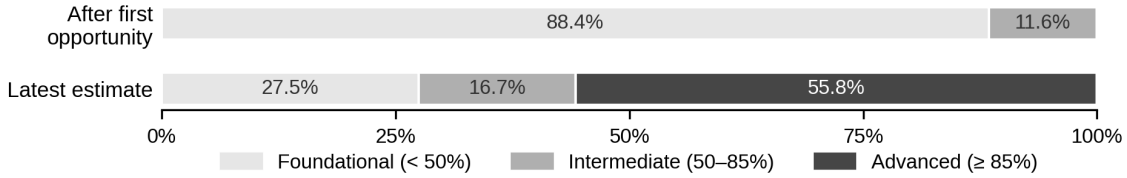
The mean estimate across the 46 started students rose from 34.5 percent to 75.3 percent, a mean paired increase of 40.7 percentage points that was statistically significant, $t(45) = 14.68$, $p < .001$, with a very large effect size of $dz = 2.16$. The increase was significant within every grade level. Every started student's mean rose, with individual changes ranging from 0.3 to 68.6 percentage points. The mean change ranged from 36.0 to 48.8 percentage points across the four grade levels. Because the study did not test for differences between grade levels, these values are reported as stratum-level descriptions rather than as evidence that any grade level responded differently to the platform.

Movement Between Knowledge Tiers

Figure 19 presents the distribution of the 3,494 student–concept pairs across the platform's knowledge tiers at the two measurement points.

Figure 19.

Movement Between Knowledge Tiers



Note. n = 3,494 student–concept pairs. Tier thresholds are those applied by the platform, namely 50 percent for the intermediate tier and 85 percent for the advanced tier.

The percentage of pairs at the advanced tier has increased from 0.0 to 55.8 percent, whereas the percentage at the foundational tier has declined from 88.4 to 27.5 percent. The intermediate share has increased slightly from 11.6 to 16.7 percent, as it is a band through which pairs pass rather than accumulate. The advanced share must start at zero, since a single opportunity cannot have an estimate above the 85 percent threshold from the initialization constant.

Practice Volume and the Magnitude of Change

Table 23 groups the started students by the size of their change and reports the volume of practice recorded in each group.

Table 23.

Practice Volume by Size of Change

Magnitude of Change	Students	Mean Change (pts)	Mean Adaptive Tests Taken	Mean Card Ratings	Mean Practice Opportunities
25.0 points or more	36	+49.0	2.6	103	294
10.0 to 24.9 points	5	+18.6	1.6	42	200
5.0 to 9.9 points	2	+7.1	1.0	18	76

Less than 5.0 points	3	+0.8	1.0	15	105

Note. n = 46 started students. Values are means within each group. Students in the lowest group practiced most concepts only once, so no change can be observed for those concepts.

The magnitude of change tracked the number of adaptive tests and card ratings recorded. Thirty-six students gained at least 25.0 percentage points, averaging 2.6 adaptive tests and 103 flashcard ratings. The five who gained between 10.0 and 24.9 points and the two who gained between 5.0 and 9.9 had taken fewer tests and recorded fewer ratings. The three whose change was below 5.0 points had each taken a single adaptive test and recorded fewer than 20 ratings, leaving most of their concepts with only one opportunity. Taken together, five of the 46 started students recorded changes below 10.0 percentage points, and three enrolled students generated no records at all, so a comparable change was not observed for every student enrolled.

Discussion

The recorded estimates grew constantly over the period, both within concepts as opportunities accumulated and within students as practice overall accumulated. The pattern in Table 21, in which the increase is largest at the second opportunity and diminishes upward along the scale, is the expected behavior of the BKT update: each observation is weighted against the current estimate, so the same evidence produces a smaller change after the estimate already has a large value. The uniform change across all 46 started students,

with a huge effect size, implies that the recorded practice was enough to move the model's estimates for every student who used the platform.

Four conditions govern interpretation. First, the estimate after the first opportunity is not a pre-test score and the design is not a pre-test and post-test comparison; a one-week usage period is too short to attribute a difference in measured performance to the platform, so this section reports the trajectory the model recorded rather than a claim of learning gain. Second, the estimate is model output under the parameters stated in Chapter II rather than a test score, and its arithmetic correctness, though not its predictive validity, was established under Statement of the Problem No. 3. Third, the model contains no forgetting term, so an estimate can move only when an opportunity is recorded, meaning an unchanged estimate indicates absence of practice rather than absence of learning. Fourth, each student is measured on the concepts they practiced, so these figures are not comparable with the coverage measure in EQ (4).

The relationship between practice volume and magnitude of change in Table 23 is consistent with the structure of the model, in which the transition parameter is applied only when an opportunity is recorded. Students in the largest-change group recorded the most adaptive tests and card ratings, while those whose concepts were practiced only once showed little or no change. Because the design includes no comparison group and does not control for differences among the students themselves, this relationship is reported as an observed association rather than as evidence that practice volume alone produced the change. The students' and educators' assessment of the platform

following this implementation period is reported under Statement of the Problem No. 5.

LEVEL OF ACCEPTABILITY OF THE PLATFORM

This section describes how the evaluators determined the quality of the platform at the end of the implementation period according to the eight characteristics of ISO/IEC 25010 specified in Chapter I: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The same model has been used in a recent Philippine evaluation of educational software that was assessed by a panel of ten expert evaluators (Alan et al., 2026).

Respondents and Instrument

A total of fifty-seven (57) respondents filled up the questionnaire using Google Forms at the end of the usage period (13 to 17 September 2026): 46 Junior High School students, one educator, and 10 information technology practitioners, who assessed the application as technical experts. The students and the educator, the number of whom coincides with the users whose performed tasks are mentioned under Statement of the Problem No. 4, evaluated only the features available on their respective portals. The statements were rated

using the four-point Likert scale provided in Table 2, which does not include a neutral option to promote a decisive response. In addition, all the 57 respondents accomplished all the close-ended statements; hence, no item was eliminated.

Each characteristic is defined by its own set of statements: two each for functional suitability, performance efficiency, compatibility, reliability, maintainability, and portability; three for security; and four for usability. Since the characteristics represent distinct attributes of software quality as opposed to facets of a single underlying trait, the 19 statements were not viewed as a unidimensional scale, and a composite reliability coefficient was not calculated for the set. The characteristics were analyzed separately using frequency count, weighted mean, standard deviation, and the verbal interpretation ranges in Table 2, with each mean rounded to two decimal places before determining its range. Because the three respondent groups evaluated the platform from different perspectives, results are reported both in aggregate and by group.

Level of Acceptability by Quality Characteristic

Table 24 presents the frequency distribution, weighted mean, standard deviation, and verbal interpretation of the 19 statements, grouped by characteristic, with the composite mean of each.

Table 24.

Acceptability Ratings by Indicator and Quality Characteristic

Indicator	Frequency of Responses	Weighted Mean	Standard Deviation	Verbal Interpretation

	4	3	2	1			
1. The app performs the tasks it is supposed to do.	28	21	5	3	3.30	0.84	Strongly Agree
2. The app gives correct and accurate results.	27	23	4	3	3.30	0.82	Strongly Agree
Composite Mean - Functional Suitability					**3.30**	**0.63**	**Strongly Agree**
3. The app responds quickly when I tap or click something.	1	5	24	27	1.65	0.72	Strongly Disagree
4. The app uses resources efficiently.	2	13	28	14	2.05	0.79	Disagree
Composite Mean - Performance Efficiency					**1.85**	**0.53**	**Disagree**
5. The app works well on my device (phone, tablet, or computer).	25	22	4	6	3.16	0.96	Agree
6. The app works smoothly without conflicting with other apps I use.	25	22	8	2	3.23	0.82	Agree
Composite Mean - Compatibility					**3.19**	**0.69**	**Agree**
7. It was easy to understand how to use the app.	28	21	6	2	3.32	0.81	Strongly Agree
8. It was easy to learn how to use the app.	24	21	9	3	3.16	0.88	Agree
9. The app was easy and comfortable to use overall.	18	25	10	4	3.00	0.89	Agree
10. The app looks good and pleasant to use.	24	23	10	0	3.25	0.74	Agree
Composite Mean - Usability					**3.18**	**0.41**	**Agree**

11. The app works without crashing or freezing unexpectedly.	26	21	4	6	3.18	0.97	Agree
12. If something goes wrong, the app recovers well and keeps my data safe.	22	23	5	7	3.05	0.99	Agree
Composite Mean - Reliability					**3.11**	**0.74**	**Agree**
13. The app keeps my personal information safe.	28	19	8	2	3.28	0.84	Strongly Agree
14. The app prevents other people from seeing or changing my data.	22	28	5	2	3.23	0.76	Agree
15. I feel my account is safe from hackers or unauthorized users.	32	18	5	2	3.40	0.80	Strongly Agree
Composite Mean - Security					**3.30**	**0.47**	**Strongly Agree**
16. If there is a problem, it seems easy to find where it went wrong.	29	20	4	4	3.30	0.89	Strongly Agree
17. The app seems easy to test and check for errors.	22	26	7	2	3.19	0.79	Agree
Composite Mean - Maintainability					**3.25**	**0.61**	**Agree**
18. The app was easy to install or access.	24	26	5	2	3.26	0.77	Strongly Agree
19. The app could easily replace other similar apps or tools I have used.	24	23	7	3	3.19	0.85	Agree
Composite Mean - Portability					**3.23**	**0.58**	**Agree**
Overall Composite Mean					**3.08**	**0.24**	**Agree**

Note. n = 57. The table shows how the 57 evaluators answered the nineteen statements, grouped by quality characteristics. 4 = Strongly Agree, 3 = Agree, 2 = Disagree, 1 = Strongly Disagree.

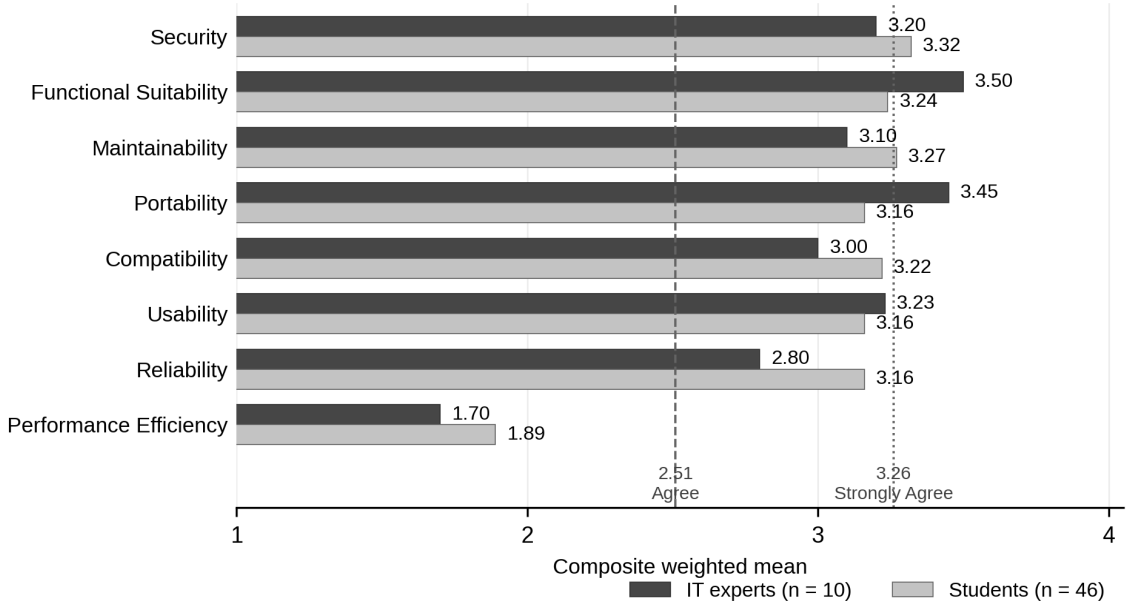
Seventeen of the 19 statements were interpreted as Agree or Strongly Agree, seven of them as Strongly Agree. The highest-rated statements were the safety of the account from unauthorized users at 3.40 and the ease of understanding how to use the platform at 3.32. Only the two performance efficiency statements fell below the agreement threshold. Response speed to a tap or click obtained the lowest mean of any statement at 1.65 (Strongly Disagree), with 51 of the 57 respondents disagreeing or strongly disagreeing, while efficient use of device resources obtained 2.05 (Disagree), with 42 disagreeing or strongly disagreeing.

Results by Respondent Group

Figure 20 compares the composite means of each characteristic between the students and the information technology practitioners, and Table 25 presents the composite means of all three groups with the aggregate result and rank.

Figure 20.

Quality Ratings by Respondent Group



Note. n = 56 (46 students and 10 practitioners). The educator is omitted as a single respondent; those values are in Table 25. The dashed line marks 2.51, the start of Agree, and the dotted line 3.26, the start of Strongly Agree.

Table 25.

Quality Ratings by Respondent Group and Rank

Quality Characteristic	Items	Students (n = 46)	Educator (n = 1)	IT Experts (n = 10)	All (n = 57)	Verbal Interpretation	Rank
Security	3	3.32	3.67	3.20	3.30	Strongly Agree	1
Functional Suitability	2	3.24	4.00	3.50	3.30	Strongly Agree	2
Maintainability	2	3.27	3.50	3.10	3.25	Agree	3

Quality Characteristic	Items	Students (n = 46)	Educator (n = 1)	IT Experts (n = 10)	All (n = 57)	Verbal Interpretation	Rank
Portability	2	3.16	4.00	3.45	3.23	Agree	4
Compatibility	2	3.22	4.00	3.00	3.19	Agree	5
Usability	4	3.16	3.50	3.23	3.18	Agree	6
Reliability	2	3.16	4.00	2.80	3.11	Agree	7
Performance Efficiency	2	1.89	1.50	1.70	1.85	Disagree	8
Overall Composite Mean	**19**	**3.08**	**3.53**	**3.03**	**3.08**	**Agree**	

Note. n = 57. Ranks use unrounded means, so Security ranks ahead of Functional Suitability although both display as 3.30.

Seven out of the eight characteristics were agreed (Agree or Strongly Agree) among the respondents. Security was the highest ranked followed by functional suitability with composite means of 3.30 and 3.30, respectively, both of which were interpreted as Strongly Agree. Maintainability, portability, compatibility, usability, and reliability were ranked next with composite means ranging from 3.11 to 3.25, which were interpreted as Agree. Performance efficiency was the only characteristic below the cut-off point, 1.85, and it ranked eighth. Overall, the composite mean of the 19 statements was 3.08 (SD = 0.24), which was interpreted as Agree, suggesting that the evaluators generally found the system acceptable.

The three groups have drawn largely consistent conclusions. Overall composite means of 3.08 for the students, 3.03 for the practitioners, and 3.53 for the single educator were produced. All three groups ranked performance

efficiency the lowest, by a considerable margin, at 1.89, 1.70, and 1.50 respectively, and evaluated every other characteristic within or above the agreement range. The differences that exist in the groups' conclusions are characteristic-specific rather than systematic: the practitioners evaluated functional suitability and portability higher than the students (3.50 and 3.45 against 3.24 and 3.16) and reliability and compatibility lower (2.80 and 3.00 against 3.16 and 3.22).

Discussion

The evaluation indicates that the target users found LearnSync AI satisfactory, with an overall composite mean of 3.08 (Agree), and seven out of eight characteristics interpreted as Agree or Strongly Agree. The two characteristics with the highest mean ratings correspond to aspects of the platform. Security corresponds to the row-level security policies defined in Chapter II, which isolate each student's data and limit educators' access to their classes. Functional suitability corresponds to the platform being able to perform the tasks it was designed for, which is consistent with the technical correctness under Statement of the Problem No. 3.

Performance efficiency was the only characteristic below the agreement threshold, and interface responsiveness received the lowest mean of any statement. This outcome is consistent with the operational context of the implementation period, during which material parsing, adaptive item generation,

and the semantic evaluation of Free Recall and Feynman responses were executed through external model calls, each adding latency experienced by the user at the time of interaction. Performance efficiency is thus identified as the primary area for improvement.

That all three groups independently ranked performance efficiency lowest reinforces this conclusion. The students and the educator used the platform on their own devices and network connections during the implementation week, while the practitioners evaluated it as a software artifact, and the three perspectives arrived at the same assessment of responsiveness.

Two considerations apply to the interpretation. First, the instrument is a structured descriptive evaluation of perceived quality, organized by the ISO/IEC 25010 characteristics, rather than a psychometric scale assessing a single latent construct. Internal-consistency coefficients across all 19 statements are therefore not an appropriate measure of its adequacy, as a respondent may plausibly rate a platform highly on security and low on performance efficiency without contradiction. The composite mean of each characteristic, reported in Table 25, is the unit of analysis, and the overall mean of the 19 statements is only a summary, not a score on a unified scale. The three groups agreeing on the same highest and lowest characteristics adds a check on consistency. Second, the evaluation measures perceived quality and is distinct from the verification of computational correctness under Statement of the Problem No. 3. Respondents rated only the features available to them, so the educator portal rests on one evaluation.

CHAPTER IV

INTRODUCTION

This capstone project is about the design, development, implementation, and evaluation of LearnSync AI: An Artificial Intelligence Based Personalized Learning Companion for Pasig Christian Academy (PCA), an innovative platform that utilizes Bayesian Knowledge Tracing (BKT) and Generative Artificial Intelligence (Generative AI) to facilitate the independent learning of Junior High School students using only educator approved materials.

The study utilized four data sources. A requirements survey was answered by 62 of the 98 students. The verification process included 12,067 BKT updates. The platform included records of 46 out of 49 sampled students for a one-week period of use. Finally, the ISO/IEC 25010 questionnaire was answered by 57 evaluators including 46 students, one educator, and 10 IT practitioners. This chapter will discuss the conclusion and address each of the research questions in turn and provide the recommendations.

CONCLUSIONS

This study aimed to determine whether LearnSync AI could address the independent study challenges of Junior High School students, compute its knowledge estimates correctly, record changes in those estimates during use,

and meet the ISO/IEC 25010 software quality standards. The following conclusions answer each research question in the order presented in the Statement of the Problem.

Current Independent Learning Practices and Challenges

The students' main challenge in studying independently is the lack of feedback and progress tracking. Twelve statements in the survey were scored as Agree or above with a composite mean of 3.01. The students' highest need was a single portal that monitors and tracks their progress while also indicating areas of improvement (composite mean = 3.38, Strongly Agree) with 85.48% desiring an all-in-one application and 87.10% wanting progress tracking. The students also indicate that their current applications only provide a final score (75.81%) and that they have no way to check their own explanations (62.90%). Mathematics (58.06%) and Science (24.19%) were identified as the most challenging areas of self-directed learning. Science was used as the initial learning content since it was the only other field that was identified across all four grade levels and would be appropriate for concept-based reviews.

Features and Functionalities Incorporated into LearnSync AI

The survey results were organized into seven system requirements, all of which were met by at least one of the platform's nine features. BKT tracks

students' knowledge of each concept based on their responses to the questions in Adaptive Flashcards and Adaptive Test Mode, an estimate that determines the difficulty level of following test questions and games. For each question type, the guess value is different: 0.05 for an identification and 0.50 for a true or false, meaning that a student's written response is more valuable than a student's selected response. Generative AI is used to extract concepts from the uploaded materials, create test questions and games, evaluate Free Recall and Feynman Technique responses, and answer student queries based on the educator-uploaded materials.

Verification of the Bayesian Knowledge Tracing Engine

The results of the calculation of the BKT engine have been verified through two means against manual calculations: 46 designed test cases and all 12,067 updates recorded. All updates were found to be within ±0.01 of the manually calculated value, resulting in an Implementation Correctness Rate of 100.00% over all categories and a maximum difference of 0.0000. As a result, the knowledge estimates given to students and used to determine difficulty levels were correctly calculated. This verifies the calculation procedure rather than testing if the estimates actually predicted student exam results.

Change in the Recorded Knowledge Estimates

Forty-six out of 49 students used the platform, providing it with 12,067 practice opportunities. Their mean knowledge estimation increased from 34.5% after the first opportunity to 75.3% at the end, which is a significant increase of 40.7 percentage points, $t(45) = 14.68$, $p < .001$, $dz = 2.16$. The increase was significant in every grade level. The proportion of concepts at an advanced level (85% and above) increased from 0.0% to 55.8%, a level the model cannot reach without correct responses. Students who practiced more gained more: the 36 students who gained 25 points or more averaged 2.6 adaptive tests and 103 flashcard ratings per student, while the three students who gained less than 5 points took only one test each. Since there was no pre-test or control group, this demonstrates the progress recorded by the platform, not proof of learning gain.

Level of Acceptability under ISO/IEC 25010

The evaluators found the LearnSync AI acceptable with an overall composite mean of 3.08 (Agree). Seven out of the eight characteristics were rated Agree or above with the highest score achieved by the Security and Functional Suitability indicators each worth 3.30 (Strongly Agree). The Performance Efficiency was the only characteristic which scored below the threshold value with a score of 1.85 (Disagree). The application's response speed was the lowest of all the 19 statements with a score of 1.65. Students, the

educator, and the IT practitioners each scored the Performance Efficiency the lowest.

Overall, LearnSync AI addresses static, unmonitored, and fragmented independent study by giving Junior High School students one adaptive platform based on their teachers' materials. Its BKT engine was verified as correct, it recorded progress for the students who used it, and its users found it acceptable, with speed as the main area to improve.

RECOMMENDATIONS

The following recommendations are based on the conclusions and the limitations of the study. They are grouped into actions the school can take to support continued use, improvements to the system, and directions for future research.

Recommendations for Action and Administrative Policy

Pasig Christian Academy should implement LearnSync AI as a home study supplement in Junior High School Science for at least a grading period, as only a week's worth of the study was observed. Other concept-based classes can be incorporated after Science. The teachers should upload their modules before a grading period begins through digital PDFs or clear, straight, well-scanned documents because all AI content is based on uploads. Students should be

oriented on the different study modes. The teachers should check the dashboard every week for concepts with less than 50% and adapt their discussions, and for students who have not logged in, as 3 of the 49 chosen students did not. Since students who practiced more recorded larger gains, the teachers should create incentives based on streaks of 3 days or bi-weekly adaptive tests. Before implementing the software on a large scale, the school should secure an AI service plan that can handle many users at once. The school must also create a policy regarding who can access student files and the duration of its retention in accordance with the Data Privacy Act of 2012.

Recommendations for System and Technical Optimization

To address the slow response in the Performance Efficiency category, tests and game contents should be generated in the background when a material was uploaded instead of while a student engages with them, and frequently-used data should be cached to avoid calling Gemini repeatedly. Furthermore, when the AI's service is unavailable, the program should automatically retry or use an alternate model or display the results it has already generated, and its response time should be logged so that speed could be objectively evaluated. Meanwhile, answers in an ongoing test should also be saved automatically so that an interruption would not impede a student's progress, and flashcards/modules should be downloadable offline. In relation to the BKT engine, a forgetting mechanic should be implemented to decrease the estimated proficiency if a

concept was not reviewed for a while, and a student's initial estimation could come from their earlier results instead of starting at 0.10 for everyone. Teachers should be able to preview the text extracted from scanned documents before it is used, and voice input could let students explain concepts out loud in the Feynman Technique. Lastly, the verification of the BKT should also be repeated every time a change is deployed to the program.

Recommendations for Future Research

The future study should have a quasi-experimental design that would include a pre-test to evaluate the impact of the platform on quarterly grades and long-term retention and explore if students' knowledge estimates match their grades on the same items. Since Mathematics was the subject students found hardest to study without a teacher (58.06%) but was outside the scope of the platform, future studies could extend LearnSync AI to Mathematics using a math engine that checks each step of a solution. Each step is either right or wrong, which fits the way the BKT model updates its estimates. The parameters of the BKT model might be tuned based on the collected data using an Expectation-Maximization algorithm or compared with the alternatives, such as Deep Knowledge Tracing and Performance Factors Analysis. Finally, the study should include the comparison of findings across different grades and different teachers, and, if the platform is going to scale to a bigger audience, analyze performance with a large number of users.

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APPENDIX A : OPERATIONAL DEFINITION OF TERMS

OPERATIONAL DEFINITION OF TERMS

Adaptive Flashcards - Digital review cards within the platform that adjust reappearance intervals (1, 3, or 7 days) and update BKT knowledge probabilities based on a student's self-reported confidence rating (Easy, Medium, or Hard).

Adaptive Test Mode - An automated assessment module that dynamically scales item difficulty (foundational, intermediate, or advanced) based on a student's real-time BKT knowledge score and calibrates scoring using format-specific guess probabilities.

Agile Scrum Framework - The iterative software engineering paradigm utilized to structure project management into distinct sprint cycles, backlogs, and rapid testing increments throughout system development.

Bayesian Knowledge Tracing - A probabilistic hidden Markov model implementing four parameters: initial Knowledge, Learning Rate, Slip Probability, and Guess Probability, to continuously track a student's knowledge of specific concepts in real time.

Change in Knowledge Estimate - The arithmetic difference between a student's final knowledge estimate and initial knowledge estimate for a given learning topic, computed at the end of the implementation period.

Cognitive Load Theory - An educational design framework that guides the platform's centralized layout to reduce extraneous mental effort caused by tool fatigue and multi-platform navigation.

Cosine Similarity - A mathematical calculation executed natively within the database using the Supabase pgvector extension to measure the semantic alignment between a student's typed Free Recall summary and the stored lesson embeddings.

Digital Badges - Gamified milestone rewards awarded to students on their progress dashboard upon reaching specific study achievements or knowledge thresholds.

Document Knowledge Score - The dynamic arithmetic average of all individual concept BKT knowledge probabilities within an uploaded document, used to display overall progress and calibrate game difficulty.

Educator Portal - The secure interface enabling authenticated teachers to upload PDF learning modules or textbook scans, assign grade-level viewing access, and monitor student knowledge analytics.

Feynman Technique - An AI-assisted study feature that guides an LLM to evaluate student explanations written in plain language by conducting jargon checks, verifying analogy use, and identifying conceptual gaps.

Final Knowledge Estimate - The Bayesian Knowledge Tracing knowledge probability recorded for a student in a given learning topic after the last practice opportunity completed within the implementation period.

Free Recall - A cognitive study mode where students type out lesson recollections without notes or prompts, evaluated automatically via semantic vector comparison.

Generative Artificial Intelligence - The artificial intelligence pipeline integrated into the platform to generate adaptive test items, flashcards, gamified content, and feedback derived exclusively from educator-uploaded materials.

Guess Probability - A parameter in Bayesian Knowledge Tracing representing the probability that a student answers an item correctly through random guessing, calibrated dynamically by question format and self-reported confidence level.

Implementation Correctness Rate - A software verification metric that quantifies the percentage of system BKT outputs that match hand-calculated expected values within a +/- 0.01 floating-point tolerance.

Initial Knowledge State - A Bayesian Knowledge Tracing parameter set to a baseline default of 0.10, representing the starting prior probability that a student has learned a concept before completing any activity.

ISO/IEC 25010 Software Quality Model - The international standard framework used to evaluate LearnSync AI across eight software quality characteristics: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Knowledge Estimate - The Bayesian Knowledge Tracing probability, denoted as P(L), that a student has learned a given concept, recomputed by the system after every practice opportunity based on the student's response.

Knowledge Threshold - The Bayesian Knowledge Tracing probability at or above which a learning topic is considered learned by a student. This study adopts a threshold of 0.85.

Learning Progress - The change and trajectory of a student's Bayesian Knowledge Tracing knowledge estimates across successive practice opportunities within the implementation period, described in terms of the initial knowledge estimate, the final knowledge estimate, the change between them, and the attainment of the knowledge threshold.

Learning Rate - A Bayesian Knowledge Tracing parameter set to a default value of 0.20, representing the probability that a student transitions from an unlearned to a learned state following a study opportunity.

LearnSync AI - The web-based personalized learning companion system designed for Junior High School students at Pasig Christian Academy.

Leitner System - A review framework integrated into Adaptive Flashcards that categorizes review items into progressive review tiers based on recall performance.

Optical Character Recognition - The image-processing technology used in the ingestion pipeline to convert scanned printed textbook pages into machine-readable text.

pgvector - An open-source PostgreSQL extension integrated into Supabase to store high-dimensional embeddings and execute vector distance calculations natively within the database tier.

PlayMe Portal - The gamified review hub featuring Crosswords, Term Matching, and Timed Trivia, where game difficulty and timer constraints are calibrated using the student's aggregated document knowledge score.

Practice Opportunity - A single scored student response within Adaptive Flashcards or Adaptive Test Mode that triggers an update of the Bayesian Knowledge Tracing knowledge probability for the associated learning topic. Interactions with AI-generated educational games, the AI Learning Assistant, and the AI-assisted study strategies do not constitute practice opportunities.

Proportionate Stratified Random Sampling - The probability sampling technique used to select 49 Junior high School student participants proportionally across Grades 7 through 10.

Retrieval Practice - A cognitive study strategy operationalized through quizzes, flashcards, and games that requires learners to reconstruct knowledge from memory rather than passively rereading text.

Row-Level Security - Database policy rules configured in PostgreSQL to isolate individual student data and restrict educator access strictly to their assigned grade-level classes.

Self-Regulated Learning - The process of self-directed study supported by platform analytics, enabling students to set goals, monitor knowledge curves, and adjust study habits without direct teacher supervision.

Slip Probability - A Bayesian Knowledge Tracing parameter set to 0.15, representing the probability that a student who has learned a concept still responds incorrectly due to carelessness, misreading, or fatigue.

Spaced Repetition - A scheduling strategy based on the Leitner system that determines when flashcards reappear for review based on prior recall difficulty.

Student Portal - The web application tier through which Junior High School students access study tools, complete adaptive quizzes, play review games, and view their personalized progress dashboards.

Study Streak - A gamification feature tracking consecutive daily study activity to foster consistent, habit-forming independent learning behavior.

Vector Embeddings - High-dimensional numerical representations of text generated from uploaded study resources, stored in PostgreSQL via pgvector for semantic search and AI retrieval.

APPENDIX B : EVALUATION SHEETS

EVALUATION SHEETS

APPENDIX C : PROJECT TIMELINE

PROJECT TIMELINE

APPENDIX D : WORK ASSIGNMENTS

WORK ASSIGNMENTS

Baruelo, John Louis E. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	The Problem and its Background	Crossword	Crossword puzzle based on the concept
1	Significance of the Study	Time Trivia	Quiz based game but with time pressure
1	Statement of the Problem	Matching Card	Matching terms and connect them with each other
2	Software Engineering Paradigm	Admin	Grade Year Level Assignments, Account Creation
2	Population and Sampling		
3	Introduction		

Leonidas, Ella Mae C. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
1	Review of Related Literature	Student Dashboard	Tracking progress, knowledge level, other info
1	Scope and Limitations	Educator Dashboard	Overall tracking progress, knowledge level of each students, other info
2	Evaluation Instrument		
2	Data Collection Method		
2	Ethical Considerations		

Soria, Marc Angelo A. : Paper and Development

Specific Tasks

Documentation Development

Chapter No	Chapter Title	Module Title	Functionality/ies
2	Introduction	Flashcards	Spaced Repetition System, creation of flashcards
2	Research Design	Feynman	Understand the concept in a simpler one, AI to generate a simpler concept
2	Architectural Framework	Free Recall	Write all of the concepts

			memorized, AI will check the written concepts
2	Algorithm	Test Mode	Overall understanding of concepts to test
2	Tool to Test for Accuracy	BKT knowledge	Knowledge calculations
2	Summary	Educator Upload	Document extraction

APPENDIX E : CERTIFICATE OF APPRECIATION

CERTIFICATE OF APPRECIATION

APPENDIX F : SOURCE CODE

SOURCE CODE

APPENDIX G : PLAGIARISM CHECK RESULTS

PLAGIARISM CHECK RESULTS

APPENDIX H : AI GENERATED CONTENT CHECK RESULTS

AI GENERATED CONTENT CHECK RESULTS

APPENDIX I : THESIS AND CAPSTONE PROJECT AGREEMENT

THESIS AND CAPSTONE PROJECT AGREEMENT

APPENDIX J : EXTENDED ABSTRACT

EXTENDED ABSTRACT

CURRICULUM VITAE